The Genesis Information System

Number Theory as the Erasure Trace of a Quantity-Free Computation

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Status and voice. This paper develops a hypothesis into a coherent, internally-consistent system; it is not a set of proofs, and it is careful never to present itself as one. The originating theory, the axioms, and the direction are Alexander Pisani's; the formalisation, the computational experiments, and the gradings were developed jointly with Claude (Anthropic). Every substantive claim carries one of three grades, marked inline where it is made. A **Fact** is a classical or newly-verified mathematical or computational result — load-bearing and checkable. A **Reading** is an interpretation of a fact in the framework's vocabulary — the contribution layer, never load-bearing for any equation. A **Lead** is a flagged conjecture or direction — explicitly not a claim. None of the classical mathematics used here is ours; the contribution is interpretive and synthetic, and is identified as such throughout.

Abstract

The logarithm $\Omega(1/n) = \ln n$ makes a single algebraic structure visible across number theory, Boolean logic, circuit topology, information theory, and the statistical mechanics of a free gas. We take this structure to be the projection of a deeper, quantity-free *genesis information system* — a substrate that has distinction, succession, memory, and computational universality, but no notion of number — and we ask how number theory arises from it. The organising hypothesis, adopted as an axiom and corroborated throughout, is thermodynamic: reversible computation leaves no permanent trace, so the only record a genesis computation deposits into its environment is the heat of its irreversible erasures, which Landauer prices at $kT\ln(\text{multiplicity})$. Its one hard pillar is that informational resistance is exactly that erasure cost — $\Omega(n) = \ln n$ is the energy, in units of kT, to forget the integer n — so the framework's central quantity has been an erasure ledger all along.

Mapping erasure systematically across substrates, we find that every irreversible operation forgets $\ln(\text{fan-in})$ nats on the maximum-entropy ensemble — the conditional entropy of its input in general — and obeys a single conservation law — the entropy chain rule, one budget with two legs — on every substrate tried; that multiplicative structure first enters at the modulo operation, where the two-legged ledger becomes a factor pair; and that the first *relational* quantity, the mutual information of two residue channels, is $\ln\gcd$, so that coprimality is exactly zero correlation. This last result reaches the Ω_{gcd} term of the resistance isomorphism's union law rather than departing from it — and, extended to banks of channels, the *entire* inclusion–exclusion of the gcd lattice is realised in the multivariate information measures. We then argue that the substrate merely *counts*: the prime basis is a natural key forced by discreteness and uniqueness, not by the geometry — which is rotation-free gauge — so primality is a property of the observer's frame, and primes are what plain

ordinals look like once filtered through uniqueness and irreversibility; and the key is recoverable from the geometry by the independence principle, the same independence that is coprimality, while a sparsity principle fails. The data structure this recovers is exactly the divisibility lattice of the Natural Operating System, obtained backward from its erasure trace. Finally, the primality signal is shown to be the von Mangoldt function via Möbius inversion — and, in new measurements, to be directly the conditional entropy of the channel $a \bmod n$ given its predecessor channels, so that a prime is a channel of pure informational novelty, the cumulative innovation of counting is the second Chebyshev function, and the Euler product, the $\ln \gcd$ coprimality, and the primality ledger stand unified as three faces of unique factorisation written in information. The individual mathematics is classical; the contribution is the axioms, the systematic reading in erasure variables, the reframe of the substrate as counting, and a small number of marked formal observations. We make no claim on the Riemann Hypothesis; we give a computational account of what primality *is* without deriving its quantitative content; and we hold the erasure reading as a corroborated consistent accounting rather than a physical fact.

Part I — The Quantity-Free Floor

1. Introduction

It has long been observed that certain operations in number theory, Boolean logic, and circuit analysis share a structure. The logarithm turns multiplication into addition; the positive integers under divisibility form a distributive lattice with gcd as meet and lcm as join; Boolean algebras are themselves distributive lattices; and Gödel's arithmetisation of syntax showed that number-theoretic operations can simulate logical computation. These correspondences are individually classical and usually treated as belonging to separate traditions. A companion result makes them a single object: under the map $\Omega(1/n) = \ln n$ — a monoid isomorphism carrying the inverse integers under multiplication into the non-negative reals under addition, unique up to scale by Cauchy's functional equation — number-theoretic operations, Boolean logic, resistance topology, and the statistical mechanics of a gas of prime modes become one algebraic structure seen from four sides. We take that isomorphism as our starting point and as our central clue.

The clue we draw from it is a hierarchy. Rather than treat the shared structure as a coincidence among finished subjects, we treat those subjects as views into one informational object, and we hold the informational layer as the deeper one: mathematical objects are studied as its projections, and whenever an argument drifts into treating quantity-native structure as ground truth, the correction is to pull back to the information layer. We call the object a *genesis information system*. The bet of the programme is that number theory, analysis, statistical mechanics, information theory, and computation are not separate subjects but views into this one system, and that number theory in particular can be watched falling out of it.

What distinguishes the genesis system is that it is *quantity-free*. It has the ingredients from which counting can be built — a primitive act of distinction, a succession that propagates that act, a memory in which configurations are held, and enough of this to be computationally universal — but it does not yet possess the notion of a number, and in particular it does not possess the primes. This is a discipline as much as a description: primality must be allowed to *emerge* from the substrate's structure, never imported into it, and any construction that quietly assumes the factorisation it is meant to produce has failed on its own terms. To keep the pre-quantitative objects distinct from their quantitative completions we carry a deliberate terminology, writing of *proto*-objects at the genesis layer whose *shadows* are the continuous, quantitative objects of ordinary mathematics — with "shadow" chosen over "projection" precisely because a shadow's kernel, what was lost in casting it, may not be fully known.

The turn this paper takes is to ask what, of such a system, could ever reach us. The answer we develop is thermodynamic. A reversible computation leaves no permanent trace — a one-to-one step can in principle be undone, and deposits nothing in its environment — so the reversible interior of the genesis system is, by construction, invisible. The only information-processing step that carries an unavoidable energy cost is *erasure*: by Landauer's principle, discarding H nats of information dissipates at least $kT \cdot H$ of heat, and any logically irreversible operation, of which erasure is the minimal and canonical instance, pays this cost. It follows that the only record a genesis computation can leave in our universe is the heat of its irreversible forgetting. The hypothesis this paper adopts as an axiom and then tests is that the structure we read as number theory is precisely that record — that arithmetic is the ledger of where the genesis computation was forced to forget.

The hypothesis has one hard pillar and one honest limit, and we state both at the outset. The pillar is a fact: the framework's central quantity, resistance, already *is* the erasure cost. Resistance equals code length, and Landauer supplies the missing leg, so $\Omega(n) = \ln n$ is the energy, in units of kT, required to forget the integer n. The erasure reading is therefore not a new construction bolted onto the framework but a recognition of what its central quantity thermodynamically has been all along. The limit is equally plain. We confirm, in a composed computation and on every substrate we map, that energy is spent only at erasures and equals the resistance erased, with reversible steps free and the accounting balancing exactly; but in simulation this is Landauer's principle applied to the distinction between a bijection and a many-to-one map, an identity true by construction. Such a test can confirm that the accounting is consistent and the erasures correctly located; it cannot *risk* the claim, because it computes the very cost it predicts rather than measuring a physical dissipation that might disagree. Genuine falsifiability belongs to the physical direction — the thermodynamics of a real gas of prime modes — which we leave parked. Accordingly the erasure reading is held throughout as a corroborated, consistent accounting and as a Lead, not as a physical fact; and because the genesis system sits, by the definition of the epistemic barrier, beyond any probe that could disconfirm it, we present the reading in information-theoretic and thermodynamic terms and do not rest it on any metaphysical claim about the nature of reality.

With the object and the hypothesis fixed, the body of the paper is a single arc. We first establish the quantity-free floor: the axioms that posit the deeper layer, the substrate of distinction and succession and memory that is provably universal, the resistance isomorphism the floor projects to, and the construction of phase — the imaginary unit itself — from the genesis operations rather than by hand (Part I). We then take the erasure turn: the axiom, its pillar, the end-to-end confirmation of the ledger as a consistent accounting, and the single organising law that every irreversible operation forgets ln(fan-in) and balances a two-legged ledger (Part II). We map that law across substrates and watch multiplicative structure enter, for the first time, at the modulo operation, where the ledger becomes a factor pair (Part III). We ask the first relational question and find that the mutual information of two residue channels is $\ln \gcd$ — coprimality exactly zero correlation — landing on the $\Omega_{\gcd}$ term of the union law and giving the framework's "innovation is connected correlation" its arithmetic face (Part IV). We then make the paper's central conceptual move: the substrate merely counts, the prime basis is forced by the discreteness of counting and the uniqueness of factorisation rather than by any geometry, primality lives in the observer's frame, and the data structure this recovers is exactly the Natural Operating System obtained backward

(Part V). We close the arithmetic arc by identifying the primality signal as the von Mangoldt function and unifying the Euler product, the coprimality result, and the primality ledger as three faces of unique factorisation (Part VI), before setting out the frontier and the honest boundary in full (Parts VII–VIII).

Two commitments govern the whole. The first is the grading already stated: Fact, Reading, and Lead are kept distinct, and no Reading or Lead is ever advanced as a Fact. The second is the honesty boundary. None of the classical mathematics here is ours — the logarithmic isomorphism, the divisibility lattice, Landauer's principle, the Shannon quantities, the Weyl clock-shift algebra, the Möbius and von Mangoldt and Euler identities, the Chinese Remainder Theorem, Rota's incidence-algebra inversion, and classical multidimensional scaling are all standard, and are attributed where used. The contribution is the axioms, the genesis framing, the systematic reading of number theory in erasure variables, the reframe of the substrate as counting, and a small number of new formal observations, each marked. We claim no new theorem, no leverage on the Riemann Hypothesis, and no derivation of the quantitative content of primality; where we give primality a computational identity we say exactly that, and no more. The discipline throughout is to refuse to claim a breakthrough where only an established fact has been found.

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2. The Axioms

The programme rests on five axioms, given here in the form they took as they drove the research — before the erasure reading of Part II was adopted. They are not all of one kind. The first three articulate what a genesis information system is and how it stands to an observer; the last two are candidate axioms, owned rather than derived, that fix how such a system's structure is costed and read. We keep them in their original form because they generated everything that follows, and because the erasure axiom introduced later joins the last two at exactly the same standing: adopted, load-bearing, and not claimed as proven. Where an axiom has a formal residue elsewhere in the corpus — a gauge degree of freedom, a theorem-form, a deflationary definition — we give that residue and lean on it, rather than on any picture of what lies behind the barrier.

Axiom 1 (The Ontological Ladder). *Reality admits a hierarchy grounded by reduction — biology to chemistry to particle physics to mathematics to the primes to information to existence itself — and at its floor reality begins undifferentiated, an OFF/ON with as yet no difference in meaning; the first act of differentiation creates the first orthogonal identity, and the act repeated unfolds the integers.* The grounding relation is not uniform along the ladder: below the mathematics line it is compositional, a whole built from parts, and above it grounding, a layer holding of a deeper one; this join is real and left explicitly marked rather than smoothed over. What the axiom fixes is the direction of explanation — the informational floor is the ground, and quantity is something that appears higher up — together with the seed of the whole construction: differentiation, applied and reapplied, is what there is to build from.

Axiom 2 (Excited information presupposes a system). *Information found in an excited state — observed, active, "on" — presupposes at least one possible system capable of exciting it; a coherent hypothetical is enough, and no unique such system is asserted.* The formal residue is a gauge. A concept can be found "on" only relative to a supporting system, and that reference contributes an observation potential which is necessary for anything to be observed at all yet cancels in every ratio, exactly as a reference voltage does. In its theorem-form the axiom is the Gelfand–Naimark–Segal construction read in the natural direction: a positive-definite correlation implies a system realising it as a covariance *(Fact, classical).* We use the axiom in its gauge and correlation forms, and note where it matters that this implication runs from positivity to system and not the other way — a direction that will constrain, rather than resolve, the deepest part of the programme.

Axiom 3 (Behind the barrier, the system is in superposition). *When the generating system lies behind an epistemic barrier it is, for the observer, in superposition over all the systems consistent with what has been observed.* This is a formal underdetermination, not a physical wavefunction. Two superpositions must be held apart: a superposition over the possible *outputs* of a fixed system, and a superposition over the possible *generators* consistent with a fixed set of observations. It is the second that the programme quantifies, and its consequences are concrete — it disqualifies any instrument that would examine the structure one piece at a time rather than all at once, and it reads the fluctuation statistics of the deepest programme as the maximum-entropy signature of an unresolved generator-superposition rather than as noise on a fixed object.

Axiom 4 (All concepts carry resistance; numbers are the quantifiable subset). *Every excited concept has a cost of being projected into the observed state — its resistance — and the numbers are exactly the part of that cost which can be laid against a shared ruler.* The resistance isomorphism measures that part, anchored at $\ln 1 = 0$; a general concept's resistance is orthogonal to the number axis and shares no ruler with it, so it is real but unmeasured rather than absent. The formal home of the unmeasured part is the imaginary direction of the resistance plane, developed in §4. This is a candidate axiom — owned, not derived — and it is stated as such: it fixes that resistance is the universal currency and that quantity is its measurable projection, without claiming this from anything more primitive.

Axiom 5 (The Probe). *Every information system carries a native probe set — the repertoire of questions it can put to a state — and actuality is relative to that repertoire: structure a native probe can read is actual, structure the dynamics conserve but no native probe can read is virtual; the conversion of virtual structure into actual structure is computation, and it is never free.* On this reading an epistemic barrier is nothing over and above the complement of a probe set, and an observer is nothing over and above a probe set holding records behind such a barrier — the two notions the earlier axioms invoked are here given a deflationary, formal definition rather than a metaphysical one. This too is owned rather than derived, at the same standing as Axiom 4. Its final clause — that conversion is never free — is the seed from which the erasure reading of Part II grows: the one unavoidable cost of making the virtual actual is the erasure of the alternatives.

Two definitions and two consequences complete the floor these axioms describe. *Excited information* is information found "on" relative to a supporting system, in the sense of Axiom 2. The ground of the whole structure is a single object seen four ways: bare existence, zero resistance, the empty message, and the most probable state of the gas of prime modes coincide — the integer 1 is $\ln 1 = 0$, the empty product, the ground state, and the unobservable identity of the isomorphism, one object under four descriptions. And quantity is not yet present at this floor: differentiation creates *directions*, not counts, so three distinctions are three orthogonal identities and cardinality is an overlay painted from an already-numbered vantage. The invariant that joins the genesis layer to the mature prime structure is therefore independence — three orthogonal proto-distinctions below, the prime axes $\{\ln p\}$ linearly independent over the rationals above — and it is this independence, not counting, that the rest of the paper watches acquire quantitative form. With the deeper layer and its floor fixed, we turn to the substrate itself: what distinction, succession, and memory build, and why that is already enough to compute anything.

3. The Genesis Substrate

The substrate has a single primitive: distinction, the act of drawing a boundary and marking one side. This is the mark of Spencer-Brown's *Laws of Form*, and in the vocabulary of ordinary logic it is negation — the one operation NOT, which toggles the marked state. Axiom 1 supplies the rest by iteration: differentiation applied and reapplied is the whole of the raw material, so if counting, memory, and universal computation are to be had at all, they must be assembled out of repeated distinction and nothing else. The task of this section is to show that they can be, and — the point that recurs throughout the paper — that quantity is not among the ingredients but among the costs.

Distinction alone does not yet count, and the reason is idempotence. The combining operations of Boolean logic satisfy $1 \vee 1 = 1$ and $1 \wedge 1 = 1$: presenting a state together with itself returns the state unchanged, and idempotence is precisely the property that forbids accumulation, since a thing combined with itself that yields back the thing can never grow. To leave the single bit and make a genuine second thing is to break idempotence — the first step from the boundary into the interior — and that step produces 2 at a cost of $\ln 2$, the genesis bit. It matters structurally that no combination available at the floor can take this step: the floor's binary operations are the Boolean joins and meets, which are idempotent, and fed the only state in existence they can only return it. What is needed is an operation that produces a new state from a single state — a *unary* operation — and so the birth of two-ness from one-ness cannot come from any Boolean combination; it can come only from a succession, a unary act that carries each state to a new one. *(The idempotence obstruction is Fact; reading the first interior step as the genesis bit at cost* $\ln 2$ *is a Reading anchored on the isomorphism of §4.)*

Before succession is built, it is worth seeing why the natural symmetries of the floor cannot supply it, because the obstruction is exact and instructive. The number line is bookended by two poles: the integer 1, which is $\ln 1 = 0$, the empty product and the bottom of the divisibility lattice, and the point 0, which under the reciprocal that carries integers into the isomorphism's domain sits at infinite resistance, the top. Two involutions act naturally on this structure — complementation $z \mapsto 1 - z$, which is NOT and which exchanges 0 and 1, and the reciprocal $z \mapsto 1/z$, which exchanges 0 and ∞. Together they permute the three special points $\{0, 1, \infty\}$, and the group they generate is the anharmonic group of the cross-ratio, of order six and isomorphic to S_3 *(Fact, classical)*. Here "three" makes its first appearance not as a count but as the order of the orbit of the floor's own symmetries — a structural three, prior to any counting of objects.

The instructive part is that this group is *finite*. Not because two involutions can never generate infinite order — in general they generate a dihedral group, which can be infinite — but because these two close: their composite $z \mapsto 1/(1 - z)$ has order three, so every product of the floor's two symmetries cycles, and the whole group has order six. A finite group contains no element of infinite order, and infinite order — the unbounded cyclic structure of $1, 2, 3, \ldots$ — is exactly what counting is. So the primitive distinction and the projective symmetries of the floor are provably insufficient on their own: **infinite order requires a carrier**, some mechanism that lifts the order-two toggle to a successor that never returns. The failure is not incidental. It is the precise reason the substrate must add structure to count, and it names what must be added.

The carrier is nesting, and succession is NOT propagated through it. Concretely, the successor has two faces. One is a *toggle*, the order-two involution flipping the current bit; the other is a *carry*, which fires when the toggle overflows and recurses to the next level of nesting depth. This is nothing other than binary increment — flip the low bit, and if it was already set, carry into the next —

and it dissolves the S_3 obstruction exactly, because the carry is the carrier the involutions lacked: nesting depth is unbounded, so the composite of toggle and carry has infinite order and generates the integers. The two faces are worth naming, since the paper returns to them: the toggle is a *phase* face, the seed of the proto-phase completed in §5, and the carry is a *depth* face, the seed of quantity and resistance. Succession welds the finite phase symmetry to an unbounded depth, and counting is the weld.

With succession in hand the memory model is forced, and it is strikingly economical: the entire state of a computation is a single integer. Writing $n = \prod_p p^{k_p}$, the integer is read as a point in the lattice of prime-exponent vectors, one axis per prime; the register addressed by the prime p holds the depth k_p; incrementing that register is multiplication by p, the map $n \mapsto p\,n$; and testing that register for emptiness is asking whether p divides n — the divisibility atom, the destructive meet-probe that is the substrate's one native test. The primes enter here in their most deflated role, as nothing more than the independent axes of the memory lattice, the register addresses along which nesting depth accumulates. They are not yet numbers in any counted sense, and nothing about their eventual arithmetic significance has been used; they are simply the directions succession can push along, which is exactly the independence that Axiom 1's closing observation identified as the invariant of the whole construction.

That this substrate is not merely suggestive but computationally universal is a theorem. A machine whose state is a single integer, whose operations are multiplication by a prime and the divisibility test — equivalently, Conway's FRACTRAN, a list of fractions acting on the integer by the first multiplication that leaves it integral — simulates the register machines of Minsky and is therefore Turing-complete *(Fact, classical).* The quantity-free substrate of distinction, succession, the integer-lattice memory, and the meet-atom already computes anything computable, with no arithmetic assumed beyond the lattice's own structure. And the manner in which universality is reached delivers the section's lesson in sharpest form. The phase face alone — the squarefree part, all exponents confined to $\{0, 1\}$, the toggles without unbounded carry — is only finite-state, a bounded symmetry that computes nothing universal. Universality enters at exactly the point where nesting depth is allowed to grow without bound, where the exponents k_p are free to run. **Quantity is the price of universality**: unbounded counting is not an ornament the substrate wears but the precise resource it must spend to compute at all, the first and most structural instance of the paper's recurring finding that quantity is paid for rather than presupposed. *(That FRACTRAN is universal and that the squarefree fragment is finite-state are Fact; reading unbounded depth as "the price of universality" is a Reading.)* One boundary must be marked to prevent a conflation: the universality established here is that of the discrete symbolic lattice machine, and it is a separate question, not settled by this result and not claimed by it, whether the analog wave hardware developed elsewhere in the corpus computes in the same sense. The two are kept apart throughout.

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4. The Resistance Isomorphism and the Proto/Shadow Discipline

The mature structure that the genesis floor projects to is a single algebraic object, and it is the object the programme takes as its central clue. Define $\Omega(1/n) = \ln n$. This is a monoid isomorphism carrying the multiplicative structure of the integers onto the additive structure of a non-negative real quantity we call resistance, and by Cauchy's functional equation — the continuous solutions of $f(xy) = f(x) + f(y)$ are exactly the scalar multiples of the logarithm — it is unique up to the choice of scale *(Fact, classical).* The logarithm is the pivot on which the four faces named in §1 turn into one: multiplication becomes addition, and number theory, Boolean logic, the topology of resistances in a circuit, and the statistical mechanics of a gas of prime modes become one structure seen from four sides. We do not re-derive the four-sided correspondence here — it is the subject of the companion paper on the isomorphism — but we take its core map as fixed and read the rest of the paper in its variables.

The isomorphism turns the memory lattice of §3 into a resistance space with a distinguished basis. An integer $n = \prod_p p^{k_p}$ has resistance $\ln n = \sum_p k_p \ln p$, a non-negative integer combination of the atomic resistances $\{\ln p\}$, one per prime; and these atoms are linearly independent over the rationals, since a nontrivial rational relation among the $\ln p$ would exponentiate to a nontrivial multiplicative relation among the primes, which unique factorisation forbids *(Fact, classical).* So the prime logarithms are an independent basis of the resistance space — the same independent axes that were the register addresses of the substrate, now seen through the logarithm as the irreducible quanta of resistance. Everything the erasure reading will later make of the primes is prefigured in this one structural fact: they are the atoms out of which every resistance is additively composed, and they are genuinely independent, sharing no common part.

Composition in series — the multiplication of integers — adds resistance, and the interaction of two integers obeys an inclusion–exclusion law worth isolating now because a later result lands on one of its terms. For any a, b,

$$\Omega_{\mathrm{lcm}(a,b)} = \Omega_a + \Omega_b - \Omega_{\gcd(a,b)}, \qquad \text{that is} \qquad \ln \mathrm{lcm}(a,b) = \ln a + \ln b - \ln \gcd(a,b),$$

which is the union law of resistance: the resistances add, and the resistance of the shared part — the greatest common divisor — is subtracted exactly once, precisely as an intersection is subtracted in the inclusion–exclusion count of a union. It is the classical identity $\mathrm{lcm}(a,b) \cdot \gcd(a,b) = ab$ written additively *(Fact, classical).* The subtracted term $\Omega_{\gcd}$ is the shared resistance of the pair, and it is exactly the quantity that the first relational measurement of Part IV will reproduce as a mutual information — so the union law is where the arithmetic of that later result already sits.

A second classical fact fixes a theme that recurs whenever π appears in this arithmetic. Two integers drawn at random are coprime with probability

$$P\big(\gcd(a,b) = 1\big) = \prod_p \left(1 - \frac{1}{p^2}\right) = \frac{1}{\zeta(2)} = \frac{6}{\pi^2},$$

the reciprocal of the value of the Riemann zeta function at 2 *(Fact, classical).* The exponent 2 is not incidental: it is the *arity* of the relation, the number of integers drawn, and it is what places the evaluation at $\zeta(2)$ and so, through $\zeta(2) = \pi^2/6$, brings in π. This is the first sight of a claim the framework makes repeatedly — that where π enters this number theory it enters through the arity of a relation rather than through any circle — and the observation is the seed of the proto-object we call proto-pi. *(That the density is $6/\pi^2$ is Fact; reading the exponent as an arity, and π as arriving through pairing rather than geometry, is a Reading.)*

This is the point to fix the terminology that keeps the two layers from being conflated, since the genesis floor and its mature shadow will be discussed together throughout. A *proto-object* is a genesis-layer structure, pre-quantitative; its *shadow* is the continuous, quantitative object of ordinary mathematics that completes it. Three such pairs recur. *Proto-pi* is the bare binary relation just met, whose quantitative completion is π, reached through the arity-two toll that fixes the evaluation at $\zeta(2)$. *Proto-phase* is the two-valued sign $\{+, -\}$ — the toggle face of the successor in §3 — whose completion, obtained by filling the sign in with the roots of unity, is continuous phase, and whose construction from the substrate is the business of §5. *Proto-e* is the integrating bridge between the multiplicative count and the additive resistance, whose completion is the base e that is the natural base of the isomorphism itself, the constant that mediates between the exponent and its logarithm. The discipline attached to these names, enforced in the glossary, is that a proto-term is retained only while it carries both a definite pre-quantitative meaning and a named completion; a proto-object with no completion, or a completion with no genesis-layer content, is dropped.

The choice of the word *shadow* over *projection* is deliberate and is an admission rather than a flourish. A projection has a known kernel: one knows exactly what is discarded in forming it. A shadow shows only a silhouette, and the object that cast it — and, more to the point, what was lost in the casting — may not be known. Because the programme does not claim to see behind the epistemic barrier of Axiom 5, "shadow" is the honest word, and "proto" and "shadow" are held as load-bearing but provisional names, kept only as long as they earn their place. The same admission has a formal home already prepared: the resistance of Axiom 4 lives in a plane, with the measurable numbers on the real axis anchored at $\ln 1 = 0$ and the resistance of a general, unmeasured concept lying orthogonal to them on the imaginary axis. The imaginary unit that names that orthogonal direction is not postulated but built from the substrate's own operations, and constructing it is the last piece of the floor, to which we now turn.

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5. Phase Manufactured from the Probe

Two debts have been incurred. Axiom 4 placed the resistance of a general concept on an imaginary axis, and §3 named a *phase* face of the successor without saying what phase is; both invoked the imaginary unit without producing it. This section pays the debts by constructing i from operations the substrate already possesses — succession and the probe — rather than importing it. The point is not only economy. Phase generated rather than imposed is what makes the later claim of Part III intelligible, where the phase built here turns out to be the physical carrier of exactly the information that an erasure destroys; a phase added by hand could not play that role, but a phase forced by the substrate's own non-commutation can.

Work in the state space reduced modulo a prime p, spanned by residues $|0\rangle, |1\rangle, \ldots, |p-1\rangle$. Succession is the cyclic shift S, acting as $S|n\rangle = |n + 1 \bmod p\rangle$ — in the standard vocabulary of finite quantum systems this is the clock-shift X. The divisibility probe, in its role as the measurement that reads a residue, has as its natural generator the conjugate clock operator Z, acting by $Z|n\rangle = \omega^n|n\rangle$ with $\omega = e^{2\pi i/p}$. It should be said at once why this choice is canonical rather than free, since the construction's force depends on nothing having been imported: any observable that separates the p residues is diagonal in the residue basis, and any such observable whose repeated application respects the cyclic structure — reading the same relation at every step of the cycle — has eigenvalues forming a cyclic group of order p, which is to say the p-th roots of unity; every probe-reading of this kind is a function of Z, so Z is not one choice among many but the generator of the only family there is. The decisive fact is then that these two operations do not commute. Stepping and then reading is not the same as reading and then stepping, and the discrepancy between the two orders is not an error term but a definite phase.

Quantitatively, the shift and the clock satisfy the Weyl braiding relation

$$ZX = \omega\, XZ, \qquad \omega = e^{2\pi i/p},$$

where ω is a primitive p-th root of unity *(Fact, classical).* This is the manufacture of phase, stated in one line: the obstruction to commuting the dynamics with the probe *is* a root of unity, forced on the substrate by the mere existence of a shift and a conjugate measurement. Nothing was added — the phase is the failure of succession and the probe to commute, and it is the completion of the proto-phase sign of §4, the two-valued mark filled in by exactly the roots of unity the Weyl relation produces. *(That the shift and clock obey $ZX = \omega XZ$ is classical; reading this braiding as the substrate generating phase from its own two operations, and identifying ω as the completion of the proto-phase mark, is the Reading.)*

At the genesis prime the construction closes on the imaginary unit itself. For $p = 2$ the phase is $\omega = e^{i\pi} = -1$: the shift X is the bit-flip, which is NOT; the clock Z is the sign on the marked state; and the Weyl relation becomes $ZX = -XZ$, the anticommutation of the Pauli algebra. The composite of the two operations then squares to minus the identity, $(XZ)^2 = -I$, so XZ is a concrete square root of -1 — the imaginary unit realised as an operator — and equivalently the square root of the substrate's one primitive, $i = \sqrt{\mathrm{NOT}}$. In commutator form the same statement reads $[S, D] \propto -i\,\sigma_y$ at the genesis bit: the imaginary unit appears as the generator that rotates succession into the probe and back. The imaginary axis that Axiom 4 posited is thus not an assumption but a consequence, and it is present already at the parity toggle, before any counting.

The genesis bit is distinguished in a second, quantitative way that binds information to disturbance. Read as a binary test — is the residue the origin or not — the probe extracts $H(1/p)$ of information, which is maximal at $p = 2$, where it is a full bit, and declines monotonically for larger primes as the origin becomes a rarer outcome. The phase the probe imparts, meanwhile, is the Weyl angle $2\pi/p$, which is likewise maximal at $p = 2$, where it is a full π — a complete sign reversal — and shrinks toward zero as p grows. So the genesis bit is at once the most informative and the most disturbing measurement the substrate admits, the sharpest instance of the complementarity between what a probe learns and what it perturbs; and information and disturbance fall away together as the prime increases *(the entropy and phase values are Fact; reading their common peak as an information–disturbance complementarity is a Reading)*. This is the first of several respects in which $p = 2$ is a distinguished point rather than merely the smallest prime, a recurrence collected and interpreted in §22.

One boundary keeps the result in proportion. The construction requires only the parity two-cycle — the toggle and its conjugate probe — and not the integer ladder above it, so the imaginary unit is available at the one-bit floor of the substrate and does not wait on the primes or on counting. With phase in hand the quantity-free floor is complete: from the single primitive of distinction the substrate has been shown to count, to remember, to compute anything computable, to project onto the mature resistance structure of the isomorphism, and now to manufacture its own phase. Nowhere in this construction has quantity been assumed; where it appears it has been paid for. We can now take the turn that organises the rest of the paper — the recognition that the one part of this system which reaches us is the heat of its forgetting.

Part II — The Erasure Turn

6. The Erasure Axiom

Part I described the genesis system as though we could inspect it, but the axioms that grant the system its interior are the same ones that deny us most of it. By Axiom 5 the epistemic barrier is nothing but the complement of our probe set, and whatever the system does reversibly leaves no trace an external probe could read, since a reversible step can in principle be undone and returns its surroundings to where they began. The interior is, in this precise sense, silent. The turn that organises the rest of the paper is to stop asking what the system is and ask instead what it is *forced to leave behind* — and to build the reading of number theory on that residue alone.

The residue is thermodynamic, and its existence is a classical fact. A logically reversible operation carries no lower bound on its energy cost: Bennett showed that computation built from reversible steps can in principle be performed with arbitrarily little dissipation, precisely because each step can be undone. A logically irreversible operation is different in kind. By Landauer's principle, discarding H nats of information dissipates at least $kT \cdot H$ of heat, and this floor cannot be evaded, because the lost information must be exported to the environment as entropy. Erasure — the resetting of a state that forgets which state it was — is the minimal and canonical irreversible operation, the one to which every other irreversible step reduces *(Fact, classical).* It follows that the only step of a genesis computation which must deposit a permanent record in its environment is an erasure, and that the reversible bulk of the computation, however elaborate, is for any outside observer thermodynamically invisible.

This is the ground for the axiom the paper adopts and then tests.

The Erasure Axiom. *The genesis information system is observable only through its irreversible erasures, and the structure we read as number theory is the permanent record those erasures leave — the ledger of where the genesis computation was forced to forget.*

It is adopted at the same standing as Axioms 4 and 5: owned, not derived. And it is the direct continuation of Axiom 5's closing clause, that the conversion of virtual structure into actual structure is never free. Where Axiom 5 fixed only that the conversion has a cost, the Erasure Axiom names that cost — it is the erasure of the alternatives that the conversion discards — and makes it the single channel through which the system reaches us. The probe of Axiom 5 supplies the mechanism directly: the destructive meet-atom of §3 records a partition and erases the remainder, so it casts its shadow exactly where it forgets, and the actual, readable structure is precisely the accounting of the forgotten alternatives. The reading to be built across Parts III to VI is that the prime and multiplicative structure of arithmetic is the shape of that accounting.

The epistemic status of the axiom must be stated as plainly as the axiom itself, because everything downstream depends on not overstating it. The Erasure Axiom is not a physical fact and is not claimed as one. It cannot be disproven, and the reason is formal rather than mystical: by Axiom 5 the genesis system lies beyond the complement of every probe we hold, so no measurement can demonstrate that our information does not originate in such a system, and the hypothesis is therefore unfalsifiable by construction. We keep the discussion in information-theoretic and thermodynamic terms throughout and rest nothing on any picture of what the system ultimately is. Accordingly the axiom is held as a **Lead** — adopted, load-bearing, and, as §8 will show, corroborated as a consistent accounting on every substrate we examine, yet never proven. What lifts it above a mere redefinition is a single fact: the central quantity of the whole framework, resistance, already *is* an erasure cost. That fact is not an assumption but a consequence of two results already in hand, and establishing it is the work of §7.

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7. Resistance Is the Erasure Cost

The Erasure Axiom would be idle if the coincidence of "resistance" and "erasure" were merely asserted, so it is worth being exact that the coincidence is forced by two results the paper already holds. The first is that resistance is a description length: the resistance $\ln n$ is, in nats, the information content of n — the amount that must be specified to single n out from among n equally-live alternatives — which is the reading of resistance as structured self-information developed in the companion work. The second is Landauer's principle, that forgetting H nats of information dissipates at least $kT \cdot H$ of heat. Composing the two, $\Omega(n) = \ln n$ is exactly the energy, in units of kT, required to forget the integer n *(Fact, given the self-information reading and Landauer)*. The framework's central quantity has therefore been an erasure cost from the beginning; the axiom recognises what resistance thermodynamically is, and does not stipulate a new identity for it. And the fit is closer than a mere composition of results, in one respect worth isolating: Cauchy's functional equation left the isomorphism free in exactly one parameter, its overall scale, and Landauer's principle fills exactly that parameter, with kT. The single degree of freedom the mathematics could not fix is the single physical constant the thermodynamics supplies — the unit of resistance is the temperature of the ledger — so the two results do not merely stack but interlock, each ending precisely where the other begins *(the scale-freedom is Fact; reading its filling by kT as an interlock rather than a coincidence is a Reading)*.

This recognition snaps the resistance plane of §4 into thermodynamic place, and fixes the roles of its two axes. The real axis — the measurable numbers, the resistance that can be laid against a ruler — is the axis that *dissipates*: to spend real resistance is to forget, and the cost is heat. The imaginary axis — phase, the direction constructed in §5 — is the axis that *stores*: phase is carried reversibly, without dissipation, and, as Part III will show in detail, it is precisely where the information that an erasure would cost is held before the erasure is paid. So the two orthogonal directions of Axiom 4's plane are the dissipative and the conservative directions of a computation, and the erasure reading assigns each its thermodynamic office — forgetting on the real axis, storage on the imaginary — with the genesis operation that converts one into the other, the merge of §11, as the moment a stored phase is turned into dissipated heat.

The atomic structure of §4 acquires, under this reading, an equally atomic thermodynamic meaning. Because the forgetting-cost of any integer decomposes as $\ln n = \sum_p k_p \ln p$, the cost is an integer combination of the atomic costs $\{\ln p\}$, and the prime is the *irreducible quantum of forgetting* — the smallest indivisible act of forgetting, one that cannot be resolved into any cheaper forgettings — while a composite's cost is inherited additively from the primes beneath it. This is the erasure-reading's restatement of the independence of the prime logarithms, and it is the seed of the claim, argued in Part V, that a prime is nothing other than an irreducible resistance. *(That $\ln n$ is an integer combination of the $\ln p$ is Fact; reading the prime as the irreducible quantum of forgetting is a Reading.)*

Even a single prime's forgetting-cost already carries internal structure the moment a probe touches it. Applying the destructive divisibility probe of §3 to a cyclic channel of girth p partitions the cost into a recorded leg and an erased leg by the entropy chain rule,

$$\ln p \;=\; \underbrace{H\!\left(\tfrac{1}{p}\right)}_{\text{recorded}} \;+\; \underbrace{\left(1 - \tfrac{1}{p}\right)\ln(p-1)}_{\text{erased}},$$

where $H(1/p)$ is the yes/no of the origin test that the probe keeps and $(1 - \frac{1}{p})\ln(p-1)$ is the identity of the non-origin residue that the destructive probe dumps *(Fact)*. At the genesis bit $p = 2$

the erased leg vanishes identically, since there is no non-origin residue left to forget once the single alternative is named, so the genesis bit's cost is pure recorded information — the unique lossless case, and the same distinction of $p = 2$ met in §5, now in thermodynamic dress. The split shown here holds for a channel of any girth, and its operational derivation from the superposed state and the destructive probe is the subject of §10; in this section it serves only to establish that the erasure cost is not structureless but has arithmetic content as soon as it is measured.

The pillar makes the axiom a recognition rather than a stipulation, but recognition is not yet corroboration, and the axiom claims considerably more than the identity of this section. It claims that the *entire* energy accounting of a genesis computation is carried by its erasures — that reversible steps are free, that every unit of dissipation sits at an erasure, and that the books balance when the computation is run end to end over arbitrary inputs. Whether that stronger claim survives an actual composed computation is a thing to be tested rather than asserted, and §8 subjects it to exactly that test, together with the two conditions under which it would fail.

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8. The Energy Ledger: A Confirmed Consistent Accounting

The stronger claim of §7 is testable, and the companion note on the carrier reading frames the test as a kill-or-promote: run a composed genesis computation, account the energy at every step, and let the claim be killed if either of two things happens — energy appears at a step that is not an erasure, or the total energy disagrees with the sum of the erasures. These are the two kill conditions, and a framework that passes both has shown its accounting to be at least self-consistent.

The computation is a lattice-machine run faithful to the memory model of §3. The state is a single integer $n = 2^a 3^b$, two registers a and b holding the exponents of the primes 2 and 3, each ranging over $0, \ldots, 5$ so that each register carries $K = 6$ values. Five FRACTRAN-style steps are applied over an ensemble of inputs, and the energy of each step is the entropy it compresses, $H_{\text{before}} - H_{\text{after}}$ — zero for a bijection, which preserves entropy, and positive for a many-to-one step, which destroys it. Four of the five steps are reversible bijections on the register pair, and one is a deliberate erasure, the reset of b to 0 that forgets the exponent of 3.

Over a uniform ensemble of inputs the ledger reads as follows.

Step	Operation	Energy (nats), uniform ensemble
1	`INC a` $(\times 2)$ — reversible	0.0000
2	`SWAP a,b` (2 3) — reversible	0.0000
3	`ADD a ← (a+b) mod K` — reversible	0.0000
4	`RESET b → 0` (forget the exponent of 3) — **erasure**	$\mathbf{1.7918 = \ln 6 = \ln K}$
5	`DEC a` $(\div 2)$ — reversible	0.0000
	Total	**1.7918**

Every reversible step costs nothing, the single erasure costs $\ln 6 = \ln K$ — exactly the resistance of the forgotten register — and the total equals that one erasure to the last digit *(Fact)*. The result is not an artifact of the uniform input. Rerun over a random, non-uniform ensemble, the four reversible steps still cost zero to within 10^{-16}, because a bijection preserves entropy for any distribution whatever, and the erasure now costs $H(b \mid a) = 1.6587$ nats — the conditional entropy of the forgotten register, whatever that happens to be — with the total once more equal to the single erasure. Both kill conditions, checked explicitly, are untriggered: the largest energy at any non-erasure step, across both ensembles, is 4.4×10^{-16}, and the discrepancy between the total and the erasure cost is 0 for the uniform ensemble and 2.2×10^{-16} for the random one. Across ensembles, then, energy appears only at the erasure and equals the resistance erased *(Fact)*. And the general-ensemble figure carries a unification worth naming: $H(b \mid a)$ is precisely the system leg $H(\text{input} \mid \text{output})$ of the two-legged ledger that §9 states as the map's conservation law — the identity of the forgotten register given what survives — so the composed run does not merely pass its own test but instantiates, in one artifact, the same conservation shape that the operation-by-operation cartography of Part III will confirm on every substrate. The end-to-end accounting and the per-operation ledger are one law seen at two scales *(the identity of the figures is Fact; "one law at two scales" is a Reading)*.

What this establishes must be stated with care, because the temptation is to read it as more than it is, and the paper's honesty turns on not doing so. The test confirms the erasure axiom's accounting *as an accounting*: in a full composed computation, over any input ensemble, the energy is located at the erasure, equals the resistance forgotten, leaves every reversible step free, and balances end to end, surviving both conditions that were set up to kill it. But it does not promote the carrier reading to a physical fact, and the reason is structural. In a simulation this test is Landauer's principle applied to the distinction between a bijection and a many-to-one map — a mathematical identity, true by construction: the reversible steps are bijections and so cost nothing by Landauer, the erasure is many-to-one and so costs the entropy it destroys by Landauer, and the program returns the very logarithm it was written to return. Nothing computed in a simulation can trigger the kill condition, because the simulation evaluates the cost the theory predicts rather than measuring a physical dissipation that could come out otherwise. So the ledger shows the framework to be a consistent, correctly-structured erasure accounting — the erasures correctly located, the reversible steps correctly free, the books balanced universally — and it shows no more than that. A genuine kill-or-promote, a test that could actually come out wrong, requires the measurement of physical heat, for which the natural vehicle is the primon gas of Part VII, and that belongs to the physical direction the paper leaves parked. The carrier reading therefore stands where §6 placed it, a **Lead** with strong consistency support, and physical falsifiability remains its open path to promotion.

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9. The Organising Principle: ln(fan-in) and the Two-Legged Ledger

With the axiom corroborated as an accounting, the remainder of the paper is cartography — the mapping of the genesis operations' erasures, one substrate at a time — and two structural facts organise the entire map. The first is a law for the size of an erasure. On the maximum-entropy ensemble — the superposition state of Axiom 3, the stance of complete ignorance about the input — every irreducible operation forgets ln(fan-in) nats, where the fan-in is the number of distinct inputs that the operation collapses onto a single output. And the law is the uniform face of something more general: on an arbitrary ensemble the erasure is the conditional entropy $H(\text{input} \mid \text{output})$, whatever the input distribution makes it, reducing to the log of the fan-in exactly when ignorance is maximal. This generality is not hypothetical — it is what the random-ensemble run of §8 verified, where the erasure cost came out as $H(b \mid a)$ rather than $\ln K$ and the books still balanced — so the fan-in law is the maximally-ignorant face of a law that holds for every distribution. The logarithm is the generic signature of Landauer and appears at every operation without exception, so it carries no information about which operation is at hand; the whole content is in the fan-in. This is the refrain of the map: the ln is fixed, and to read an operation is to read the profile of its fan-in. *(That erasure equals the log of the fan-in on the uniform ensemble, and the conditional entropy in general, is Fact; reading the* ln *as generic and the fan-in as the content is a Reading.)*

The second fact is a conservation law for the shape of an erasure. Every deterministic operation, reversible or not, satisfies the entropy chain rule,

$$H(\text{input} \mid \text{output}) + H(\text{output}) = H(\text{input}),$$

and the erasure reading gives this two physical legs. The first leg, $H(\text{input} \mid \text{output})$, is the *system's* erasure — what the operation itself forgets, the identity of the input that the output no longer distinguishes. The second leg, $H(\text{output})$, is the *demon's* reset — the cost, in the sense of Maxwell's demon and Landauer, of recording the output and clearing that record so the operation can run again. Their sum is the whole of the input information, exactly *(Fact)*. Every operation therefore divides its input information into a part it forgets and a part that must be recorded and cleared, and the two parts always sum to the total. This two-legged ledger is the structural through-line of Part III: it holds on every substrate examined, and the only thing that changes from one substrate to the next is the multiplicity that sits inside the two logarithms.

One observation binds the map to Part I and closes the section. Because $\ln n = \Omega(1/n)$, every erasure value the cartography produces is a resistance value — the fan-in's logarithm is the resistance of the fan-in — so the erasure ledger and the resistance isomorphism are not two structures set side by side but one structure written in two vocabularies. To catalogue the erasures of the genesis operations is to read the resistance isomorphism operation by operation, in thermodynamic variables. With the law and the ledger fixed, Part III maps them across the substrates in turn — the succession channel, the additive merge and its reverse, and the wider catalogue of operations — and watches for the point at which multiplicative structure, absent from every additive ledger, first enters.

Part III — The Ledger Machinery

10. The Succession Channel and the Two-Legged Stream

The cartography begins with the simplest structure succession can build alone. Run the successor of §3 — NOT propagated through nesting — until it returns to its origin, and the result is a *cycle*: the only finite structure a lone succession admits. The number of steps to return, the channel's *girth* g, is a structural return-time, not a quantity imposed from outside — it is read off by running the dynamics and watching for the origin, in the manner of a stride-count rather than a measurement. The quantity-free discipline is nonetheless stated at the door, because it will be tested before the section ends: we do not claim the genesis substrate houses the integers. The integer girth is scaffolding, the channel family is indexed by $\mathbb{Z}/g$ for our convenience, and what the analysis keeps is not g but the resistance $\ln g$ and the erasure-decomposition that rides on it.

Two ingredients complete the model, both already in hand. The state of ignorance about the channel is the superposition of Axiom 3 in its state form — the maximum-entropy belief over the g internal positions, uniform, of entropy $\ln g$. And the instrument is the substrate's one native test, the destructive meet-atom of §3, here the binary question *is the state the origin?* Being destructive, the probe records the partition it induces and erases everything else.

Applied to the superposed channel, the probe splits the resistance with no slack, and the split is the chain rule of §7 in its operational home:

$$\ln g \;=\; \underbrace{H\!\left(\tfrac{1}{g}\right)}_{\text{recorded}} \;+\; \underbrace{\left(1-\tfrac{1}{g}\right)\ln(g-1)}_{\text{erased}}.$$

The recorded leg is the yes/no of the origin test; the erased leg is the *which-stride* information of the non-origin class — $g-1$ equiprobable positions, entropy $\ln(g-1)$, carried with weight $(1-\frac{1}{g})$ — which the destructive probe coarsens away. Numerically the identity closes to machine precision for every girth tried *(Fact).* One qualification keeps the thermodynamics honest: the erased leg is genuine heat only because the probe is destructive; a non-destructive probe would retain the residue hidden rather than erase it, and would pay nothing yet.

The channel also resolves a refinement the direction of the programme had left open — whether the processing erasure and the probe erasure are two independent costs that happen to sum to $\ln g$, or two faces of one conserved quantity. The answer is the second, and it can be read off three processes run on the same channel. Reversible succession, the bare tick, records nothing, dumps nothing, and retains the full $\ln g$ reversibly; the destructive read records $H(1/g)$, dumps $(1-\frac{1}{g})\ln(g-1)$, and retains nothing; the overwrite that resets the clock records nothing and dumps the whole $\ln g$. Every row totals $\ln g$ exactly — at $g = 6$, for instance, the probe's $0.45056 + 1.34120$ and the overwrite's 1.79176 are the same number reached by two routes *(Fact).* So $\ln g$ is *one budget with two faces*: a single conserved state-information that any process partitions into recorded, dumped, and retained, never two addends stacked. *(The conservation is Fact; the resolution of the processing/probe relation is a well-grounded Reading.)*

At the genesis bit the split degenerates, and in the telling direction. At $g = 2$ the erased leg is $(1-\frac{1}{2})\ln 1 = 0$ identically: the probe records a full bit and coarsens nothing, the unique lossless channel — the same distinction of $p = 2$ that §5 found from the algebra side and §7 met in thermodynamic dress, now appearing operationally. From that anchor the extraction efficiency $H(1/g)/\ln g$ falls monotonically — 1.000 at $g = 2$, then 0.579, 0.311, 0.211, 0.127 along the primes — and the erased fraction climbs toward one. Under the genesis probe, the resistance of a larger channel is increasingly *erased* rather than recorded: the first quantitative texture beneath the central hypothesis, resistance as erasure-dominated with the genesis bit as its lossless anchor. *(The fractions are Fact; the erasure-domination reading is a Reading.)*

Before the stream is built, a correction must be recorded, because this substrate is where an earlier claim of the programme failed and was graded down, and the failure is instructive. The structural test *does the cycle have a proper sub-period?* — run genesis-natively, by stepping succession at every stride and watching for an early return, with no arithmetic factoring anywhere — returns exactly the primes: $2, 3, 5, 7, 11, 13, \ldots, 47$ over the girths tried. The test is faithful. But it is circular as a derivation, because indexing the channels by $\mathbb{Z}/g$ put the divisor structure in at the door: a proper sub-cycle of the g-cycle *is* a proper divisor of g, so the uniqueness of the prime channels was moved across from the scaffolding, not produced by the dynamics. The claim "primality as an output," made in an earlier stage of this work, is accordingly corrected to **primality as a faithful relabel**, and the discipline going forward is the one this section has practiced: accept the stride-skeleton, study the erasure — for the erasure accounting itself is non-circular, holding for every g, prime or composite, and presupposing no factorisation. Deriving the integers from erasure, making elements earn their existence through the ledger alone, remains an open and harder target, and nothing in this section achieves it. *(The correction is load-bearing for the paper's honesty and recurs at §13 and §14, where the same circularity re-enters by other doors.)*

The single snapshot, however, saw only one leg of the machinery, and the channel's full ledger appears when it is made to *emit*. Let a bank of clocks $\{g_1, \ldots, g_N\}$ tick off one shared succession, and let the probe read one bit per clock — does clock i sit at its origin at step t — so that the system emits a bitstring $B(t)$ from its current state. Over the joint period $L = \mathrm{lcm}(g_1, \ldots, g_N)$ a demon reading the stream must, to read repeatedly, reset its record between readouts, and the reset is where the second interface of erasure enters. Per read-reset cycle,

$$\underbrace{\big(\ln L - H(B)\big)}_{\text{coarsening — the observation}} \quad + \quad \underbrace{H(B)}_{\text{reset — the demon's bill}} \quad = \; \ln L,$$

exactly: the coarsening leg is the destructive measurement discarding what the bitstring cannot resolve, and the reset leg is the recorded string cleared so the demon can read again *(Fact; the identification of the two legs with the programme's two erasure interfaces is a Reading)*. The kept leg of the earlier snapshot is thereby exposed as **deferred erasure** — free-looking at the moment of recording, paid back as heat at the reset — and over a full operating cycle the demon dumps the entire $\ln L$. What looked, one probe at a time, like information extracted without cost was a debt scheduled for the reset.

The stream also delivers the map's first *composition law*, and it is the framework's own. The recorded information — the demon's bill — decomposes as $H(B) = \sum_i H_i -$ redundancy, the sum of the per-channel marginals less what the channels share, which is the connected correlation of the bank. Measured across banks, the redundancy is exactly zero precisely when the bank is pairwise coprime: the bank $(2, 3)$ records $H(B) = 1.330 = \sum_i H_i$ with zero redundancy, and $(2, 3, 5)$ and $(3, 5, 7)$ likewise; while $(2, 4)$ carries redundancy 0.216 and $(2, 4, 8)$ carries 0.419, their bits correlated through the shared factor *(Fact, from the Chinese Remainder Theorem and the entropy accounting)*. The principle the programme states abstractly as *innovation is connected correlation* here acquires its ledger form: independent channels record independent information and pay full price at the reset; channels sharing structure record less and pay less, the demon's discount equal exactly to the connected correlation. Sharing structure buys a cheaper Maxwell's-demon bill at the cost of carrying less information *(the discount reading is a Reading)*. This is the first appearance in the cartography of coprimality as a *thermodynamic condition* — additivity of the reset cost — and it is the direct ancestor of the relational result of Part IV, where the same shared structure is measured as a mutual information and lands on $\ln \gcd$.

One more reading falls out of the bank comparison, and it points at a parked lead. Any bank containing the genesis bit gets $H(1/2) = 0.693$ of recorded signal from that channel with zero coarsening; drop it for larger primes and the economics invert — the bank $(3, 5, 7)$ records *less* than $(2, 3, 5)$, 1.547 against 1.830 nats, yet costs far more, 4.654 against 3.401, almost all of it coarsening. Information is cheapest to generate at the genesis bit, which is why a single bit should be the easiest thing such a system emits *(the numbers are Fact; the "cheapest source" reading connects to the observation-probability lead parked in Part VII)*.

The honest boundary of the substrate closes the section and hands the map its next task. The *system* here is still reversible — the clocks only tick, and every erasure in the stream ledger is the demon's, the record reset of the Maxwell's-demon form, not the dynamics' own. Nothing in this section has yet made the genesis system itself forget. The next substrate does exactly that: an operation in which the system's own dynamics are irreversible, so that the system pays alongside the demon — the additive merge.

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11. The Additive Merge: Heat, Storage, and the Reverse

The system's first irreversible operation acts on the barest objects the substrate affords: *proto-integers*, unary tallies that carry length and nothing else — no multiplicity structure, no factorisation, nothing a number has beyond extent. The merge takes two tallies of lengths a and b and joins them into one of length $n = a + b$, and in doing so forgets the split: the output remembers its length but not which pair produced it. The forward map is many-to-one, and for the first time in the cartography it is the *dynamics* that pay, not the demon.

The cost obeys the fan-in law of §9 with a multiplicity worth counting carefully. A merge whose output has length m forgets which ordered split (a, b) with $a + b = m$ produced it; with $a, b \geq 0$ there are $m + 1$ such splits, uniform under maximum ignorance, so the merge-erasure is

$$\varepsilon(m) = \ln(m + 1).$$

The processing erasure grows logarithmically with the proto-integer built *(the split-count and its logarithm are Fact; reading it as the processing-erasure law is a Reading)*. And the law is robust in a way the thermodynamic reading requires: under both a bounded-uniform and a geometric input prior, the splits conditional on a fixed sum come out uniform — the prior's weights cancel in the conditional — so $\varepsilon(m)$ depends only on the output's size and never on the input distribution, a temperature-independence confirmed under both priors. The realisation factor $R = m + 1 = e^{\varepsilon(m)}$ likewise reproduces, on bare tallies, the multiplicity count that the corpus's gas test had found on prime-encoded registers — a check flagged in advance of the experiment, and landed *(both Fact)*.

One point of hygiene on the correspondence, small in appearance and load-bearing in substance. The value $\ln(m + 1)$ equals the resistance of the inverse integer $1/(m + 1)$ — but these are two different things sharing one number. The merged multiplicity is $m+1$, not m: the off-by-one is the marker that the proto-layer quantity is the *split count*, not the proto-integer wearing a resistance hat. The ln hinge is the same; the object inside it is not. The discipline of §9 — the log is generic, the multiplicity is the content — is precisely what makes this distinction visible rather than a pedantry, and ignoring it is how layer-conflation starts.

The merge closes its own two-legged ledger, and this is the substrate where the ledger's two legs first belong to two different parties. Over one merge-read-reset cycle, the system's merge-erasure plus the demon's reset on the recorded output sum to the whole input information,

$$\underbrace{H(\text{split} \mid \text{sum})}_{\text{system — merge}} + \underbrace{H(\text{sum})}_{\text{demon — reset}} = H(a, b) = H(a) + H(b),$$

confirmed exactly under both priors — 4.3944 nats for the uniform prior on $\{0..8\}$ and 5.0040 for the geometric prior of mean four, the ledger balancing to the last digit in each case *(Fact)*. It is the chain rule $H(a, b) = H(n) + H(\text{split} \mid n)$ realised as a division of labour: the same conservation shape the stream had, but with the system now paying its own leg alongside the demon's *(the reading of the two legs as two payers is a Reading)*.

Storage gives the merge a thermodynamics of *history*, and it is here that the substrate first shows a structure richer than a single operation's cost. A stored intermediate is a bare proto-integer — it remembers its length, not its composition — so every reuse of a stored tally re-forgets a split, and a build of N through intermediate merge-outputs $m_1, \ldots, m_k$ costs $\sum_i \ln(m_i + 1) = \ln \prod_i(m_i + 1)$. Building the same N by different routes then gives very different totals: at $N = 8$, a single merge of $4+4$ costs $2.1972 = \ln 9$, a balanced binary tree over eight ones costs 9.8105, and a linear chain over eight ones costs 12.1087. **Build-erasure is a path function, not a state function** — heat, not energy: the total depends on the route taken, not merely on the endpoint reached *(the sums are Fact; the path-function reading is a well-grounded Reading).* Three trends read directly off the product form, all confirmed numerically. The floor is the single merge, $\ln(N+1)$ — the cheapest route to any N, with every finer build costing strictly more, so that build-erasure decomposes into a state-function floor plus a path excess which is exactly the thermodynamic price of storing and reusing intermediates. More merges, and larger intermediates, cost more, each piece beyond the first multiplying in a factor $(m_i+1) > 1$. And the minimiser is smallest-first: combining the smallest pieces first keeps every intermediate — and therefore the erasure — lowest, the product-minimising shape familiar from optimal-merge ordering *(the ordering is Fact; the link to the Huffman shape is a Reading).* Nor is repetition special: at three pieces building 6, the all-equal $2+2+2$ costs 3.5553, *more* than the mixed $1+2+3$ taken smallest-first at 3.3322 and $1+1+4$ smallest-first at 3.0445 — what sets the cost is the profile of partial sums, not whether the pieces are equal *(Fact).*

The most consequential face of the merge, however, is its reverse. Forward, $(a, b) \to n$ forgets the split; force-reversing is therefore one-to-many, and a system driven backward through the merge with no energy supplied scatters uniformly over all $n+1$ ordered splits — a spread whose entropy is exactly $\ln(n+1)$, the amount erased. The reverse-scattering *is* the erased information made manifest, laid out as a distribution over configurations; collapsing it back to the original split requires injecting the same $\ln(n+1)$ that the forward merge dumped *(Fact).* What was heat in the forward direction is configuration-space spread in the reverse, one quantity in two appearances.

And the configuration space the scattering lives on is not structureless. The $(n+1)$-dimensional preimage space, equipped with the shift X that steps the split index and the clock Z that phases it with $\omega = e^{2\pi i/(n+1)}$, satisfies the Weyl braiding relation $ZX = \omega XZ$, verified for $n = 1$ through 5; and at $n = 1$ — the genesis bit, two preimages, $\ln 2$ erased — the algebra is exactly the Pauli algebra, with $X = \sigma_x = \mathrm{NOT}$, $Z = \sigma_z$, and $\sqrt{\mathrm{NOT}}$ present with its characteristic i *(Fact).* Three things thereby line up at the number two: the genesis bit of the succession channel, the two preimages of the merge to length one, and the Pauli/$\sqrt{\mathrm{NOT}}$ algebra that §5 constructed independently from succession and the probe — a lining-up that is precise but is, at this point, noted rather than explained *(Reading; it is collected with the other recurrences of the genesis bit in §22).*

The section's closing observation is the one that sets up the next, and it must be stated with its negative half intact. The scattering is *not* the Weyl structure — it is its **phase-blind shadow**. The states $|+\rangle$ and $|-\rangle$ on the two-preimage space carry identical classical probabilities yet are orthogonal, and superposing them cancels one outcome to zero amplitude — an interference that no mixture of probabilities can produce, as the verification exhibits directly: mixing $(\frac{1}{2}, \frac{1}{2})$ with $(\frac{1}{2}, \frac{1}{2})$ returns $(\frac{1}{2}, \frac{1}{2})$, while the amplitudes cancel. The scattering supplies the probabilities; the phase — the i — is what it lacks, and phase is precisely what enables interference *(Fact).* Even the path-structure of the storage results points the same way: the build cost is non-associative — grouping matters — even though the sum is associative, and order-sensitivity of exactly this kind, laid over amplitudes rather than probabilities, is what interference *is*. Whether the genesis dynamics supply the missing phase is not left as a speculation; it is the question the next section answers.

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12. Phase on the Merge: Generated, and Its Cost

The temptation at the end of §11 is to paint the phases on by hand — to decorate the scattering with the clock phases and declare the Weyl structure complete. The discipline of the paper forbids it, and the substrate turns out not to need it. The question actually asked was whether the two genesis operations already in hand *generate* the phase on the scattering's own space, and the answer is that they do.

Take the preimage space of the merge to n and place on it the two operations the substrate owns: S, the succession-shift on the split index, $|k\rangle \mapsto |k+1 \bmod (n+1)\rangle$ — the Weyl X — and D, the origin meet-atom probe, the projector $|0\rangle\langle 0|$. They fail to commute, and the algebra they generate contains the Weyl clock Z — the phase operator itself — for every n tried, $n = 1$ through 5: conjugating the probe by powers of the shift produces every diagonal projector, $|k\rangle\langle k| = S^k D S^{-k}$, and these span all diagonal operators including $Z = \sum_k \omega^k |k\rangle\langle k|$ *(Fact)*. At the genesis bit the statement sharpens to an identity: the commutator is the phase generator,

$$[S, D] = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i\,\sigma_y,$$

the very i of $\sqrt{\text{NOT}}$, produced by the substrate's own non-commutation *(Fact)*. This is the construction of §5 re-situated exactly where the merge needs it — on the reverse-scattering's configuration space — and generalised: the phase is available at every n, not only at the prime channels of the earlier construction. One limit is stated at once, because the grading requires it: what is established is that the phase is *structurally available* — the generating operators are present and produce Z. It is not shown that the merge dynamics evolve any particular phased state; that remains an open Lead, and nothing below assumes it closed.

With the phase in hand its price can be measured, and the price is the section's central result. Put a definite phase on the scattering — the coherent superposition of the $n+1$ preimages with the clock phases — and read the cost through decoherence. The classical scattering is the maximally mixed state on the preimage space, entropy $\ln(n+1)$; the coherent phased superposition is a pure state, entropy zero; and decohering the pure state — losing the phase — creates exactly $\ln(n+1)$ of entropy. The match is exact at every n tried: 0.6931 at $n = 1$, 1.0986 at $n = 2$, 1.6094 at $n = 4$, 2.1972 at $n = 8$, each equal to the merge-erasure to the last digit *(Fact)*. **The phase cost is the merge-erasure.** Phase is not a separate expense stacked on top of the merge; it is the same $\ln(n+1)$, seen as the coherence whose loss is the merge's irreversibility. The additive merge is irreversible precisely *because* it decoheres — a phase-preserving merge would be reversible, zero erasure, the split recoverable from the phase — so the erased split does not vanish into nothing: **phase is the carrier of the erased information**, and the heat of the merge is the death of its phase *(the entropies are Fact; the identification of merge-irreversibility with decoherence, with phase as the carrier of the erased split, is a well-grounded Reading resting on standard decoherence-as-erasure)*. A new experiment, run for this paper, makes the recoverability *operational* rather than entropic: encode the split k as the coherent clock state $|\psi_k\rangle = d^{-1/2} \sum_j \omega^{jk} |j\rangle$ and measure in the conjugate Fourier basis — the split is recovered with probability exactly 1, for every k, at every n tried; decohere first, and the best achievable recovery is exactly chance, $1/d$ *(Fact)*. Recovery certain with the phase, chance without it, and the entropy created in between is the merge-erasure: the reversibility claim is now a demonstrated protocol. This is the promise of §5 and §7 discharged: the imaginary axis stores what the real axis would dissipate, and the merge is the substrate's own demonstration of the conversion.

The last measurement of the section is a deliberate test of the substrate's limits, and its negative result is as informative as the positives. Does the generated phase let the ledger tell one split of a given n from another? Two parts, with opposite answers. The per-split phase *angle* varies with the split index — at $n = 8$ the splits step through 40° increments, $(1, 7)$ at 40°, $(3, 5)$ at 120° — so splits are now distinguishable *by phase*, which the bare scattering could not do. But the per-split phase *cost* is $\ln 9$ for every split without exception — uniform, exactly as split-blind as the merge-erasure itself — and the angle is *positional*, linear in the index k, not arithmetic: nothing in it marks any split as special beyond its position in the cycle *(Fact)*. So the additive system distinguishes splits by a positional label while costing them identically, and cannot single out any arithmetically-distinguished split; making the ledger sensitive to *which* numbers compose a sum would require multiplicative structure, which a purely additive proto-integer system does not contain. The system is incomplete in exactly the anticipated way — an expected limitation rather than a failure, since primality is absent from a purely additive world by construction — and the incompleteness is a signpost: the cartography must find the operation through which multiplicative structure enters. That operation is found in the catalogue of the next section, and it is division.

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13. The Operation Catalogue and the Multiplicative Gateway

With the law of §9 in hand — the ln is generic, the fan-in is the content — mapping an operation reduces to counting its fan-in, and the cartography can be extended across the substrate's discrete operations systematically. The catalogue that results has three families, and the manner in which the third differs from the first two is the discovery of the section.

The additive batch comes first, on proto-integers $0..M$ under the uniform prior (the tables quoted take $M = 8$). The merge, already mapped, has fan-in $n + 1$ at output n. Projection, $(a, b) \mapsto a$, forgets the register b entirely, with fan-in $M + 1$ at every output — cost $\ln 9$ regardless of the value kept; reset, $a \mapsto 0$, likewise forgets its register at flat fan-in $M + 1$, and is the limiting case in which everything is forgotten and nothing recorded, so its demon leg is zero. Max has fan-in $2m + 1$ at output m — a larger maximum forgot more — and min is its exact mirror at $2(M - m) + 1$. The family thus splits into *flat* erasers, whose fan-in is constant and whose cost is independent of the value (projection, reset), and *graded* erasers, whose fan-in varies with the output (merge, max, min); and the two-legged ledger closes for every one of them, the system leg plus the demon leg returning the input information exactly — 4.3944 nats for the two-register operations, 2.1972 for reset — in every row *(the counts, the flat/graded split, and the ledger closures are Fact; reading the fan-in profile as the operation's signature, with the* ln *its common wrapper, is a Reading)*.

The string-rewriting family widens the catalogue beyond arithmetic altogether, and contributes a structure the additive family cannot show. On binary strings of length $L = 5$, take the pattern $B = \texttt{11}$ and act at its first occurrence: *truncate* cuts the string there and keeps the prefix, *delete* splices the pattern out, *substitute* replaces it. The novelty is that the erasure is **conditional** — gated by the match, so that the test and the forgetting are fused into one act. Of the 32 inputs, the 13 containing no occurrence of B pass through unchanged, fan-in one, zero erasure; the other 19 are erased. Per-input cost is therefore bimodal — zero or positive — and the *expected* cost depends on how often the pattern occurs in the input ensemble, a dependence with no analogue among the additive operations. Fan-in ranks the family by how much it discards: truncate forgets the entire suffix from the match onward and collapses exponentially — the empty prefix receives $8 = 2^3$ inputs, cost $\ln 8$, each shorter prefix freeing more suffix — while delete forgets only *where* the pattern was and substitute only *that* it was the pattern, their maximal fan-ins bounded at 4; the mean costs rank $0.7798 > 0.5689 > 0.4496$ accordingly. And every row totals $L \ln 2 = 3.4657$, the input information, exactly: a Turing-complete rewriting substrate, with no quantum shadow anywhere in it, and the two-legged ledger passes through unchanged *(the conditional structure, the ranking, the exponential profile, and the totals are Fact; "test and forget fused" is a Reading).*

Then modulo — and the catalogue changes character. The operation $a \mapsto a \bmod m$ on inputs $0..M$ keeps the remainder and forgets the quotient $\lfloor a/m \rfloor$, and its fan-in per residue is the quotient $(M+1)/m$. Taking $M+1 = 12$ for the sake of its divisors: at $m = 1$ the fan-in is 12 and the operation is the reset; at $m = 2, 3, 4, 6$ the fan-ins are $6, 4, 3, 2$ with costs $\ln 6, \ln 4, \ln 3, \ln 2$, each residue-uniform; at $m = 12$ the fan-in is 1 and the operation is the identity; and for m not dividing 12 the fan-in is near-uniform at $\lfloor 12/m \rfloor$ or $\lceil 12/m \rceil$ *(Fact).* Two features of this row of the catalogue are unprecedented in everything above it. First, **the fan-in is a division** — every earlier fan-in was a count or a sum; modulo's is a quotient, the first division in the map, and it interpolates the whole catalogue's extremes in one parameter, from reset at $m = 1$ to identity at $m = M+1$. Second, **the ledger becomes a factorisation.** For $m \mid (M+1)$ the two legs are

$$\underbrace{\ln \frac{M+1}{m}}_{\text{forgotten — the quotient}} \quad + \quad \underbrace{\ln m}_{\text{recorded — the residue}} \quad = \ln(M+1),$$

with $(M+1)/m$ and m a *factor pair* of $M+1$. Every earlier ledger in the map split its total additively; modulo splits $\ln(M+1)$ along a factorisation. This is the multiplicative gateway, and it is visible in the ledger itself: the crossing from additive to multiplicative structure is not asserted but read directly off the shape of the two legs *(the factor-pair ledger is Fact; the gateway reading is a Reading).*

The gateway also closes a loop with the beginning of the map. The value $a \bmod m$ is the position on the m-cycle — the succession channel of girth m from §10 — so modulo m is the *full* readout of the m-clock, where the Stage-1 probe read only the single bit "at origin?"; the quotient it forgets is how many complete m-cycles fit the range. The channel that opened the cartography and the operation that ends it are one object at two resolutions.

And the gateway carries its warning, which is the §10 circularity arriving by a second door. The clean factor-pair split occurs exactly when $m \mid (M+1)$, so the pattern of clean-splitting m across all m *is* the divisor structure of $M+1$ — a prime $M+1$ admits only the trivial pair — and it is tempting to read primality off which moduli split cleanly. But that reading *imports* the divisor structure through the indexing, exactly as the sub-period test did in §10. What the gateway legitimately locates is *where* the multiplicative and prime structure lives — in the divisibility pattern across all m, the channel-bank of the stream, not in any single modulus — while the standing problem, primality as an output of a structural test rather than of the indexing, remains open here as it was there. The erasure values themselves stay non-circular throughout *(the divisibility–cleanness equivalence is Fact; the circularity caution is the methodological discipline of §10, recurring).*

The catalogue's through-line is then stated in one sentence, because it has now been verified on every substrate the map contains: across the additive batch, the string-rewriting family, and modulo — arithmetic and non-arithmetic, unconditional and conditional, flat and graded — **the two-legged ledger is universal.** System erasure plus demon reset equals the input information exactly, in every case; nothing about an operation alters the structure of the ledger, only the multiplicity inside the logarithms; and at modulo that multiplicity becomes, for the first time, a factor pair

(Fact). The map is complete enough to ask a new kind of question. Every measurement so far has interrogated one operation at a time — one channel, one merge, one modulus. The multiplicative structure, the gateway has just shown, lives not in any single operation but in the *pattern across* them. The next question is therefore relational: what do two channels share?

Part IV — The Relational Core

14. Channel Correlation Is ln gcd

The question is posed in the map's own terms. Run two modulo channels on the same input — $a \bmod m_1$ and $a \bmod m_2$, with a uniform over one full joint period $N = \mathrm{lcm}(m_1, m_2)$ — and measure what they share: the mutual information between the two readouts. Every quantity in the cartography so far has been a function of a single integer, an ln(integer) on one rung of the resistance ladder; this is the first *two-variable* measurement, and its value is the first relational arithmetic function the ledger produces.

The result is exact:

$$I\big(a \bmod m_1\,;\, a \bmod m_2\big) \;=\; \ln \gcd(m_1, m_2),$$

confirmed to machine precision — the maximal deviation across all pairs 2..8 is 2.2×10^{-16}, re-verified for this paper — and the Chinese Remainder structure makes it an identity rather than an approximation: each channel determines $a \bmod \gcd$ and nothing more of the other's content, so $I = \ln m_1 + \ln m_2 - \ln \mathrm{lcm} = \ln \gcd$ *(Fact).* Two consequences of the formula carry the arithmetic. First, **coprimality is exactly zero correlation**: the mutual information vanishes precisely when $\gcd(m_1, m_2) = 1$, so the coprime pairs of the gcd matrix — $(2, 3), (3, 4), (5, 7), \ldots$ — are the pairs whose channels record independent information, while $(2, 4)$ shares $\ln 2$, $(3, 6)$ shares $\ln 3$, and in general **two number-channels share information if and only if they share a prime, and they share exactly** ln(**shared part**) **of it** *(Fact).* The multiplicative core of arithmetic — which numbers share which primes — has become a thermodynamic signal, readable as a correlation. This is the stream's composition law of §10 sharpened from a condition to a formula: there, coprimality gave additivity of the demon's bill; here, the departure from additivity is measured, and it is the logarithm of the shared factor.

The second consequence is where the result lands, and it is the paper's clearest instance of the framework closing on itself. The value $\ln \gcd$ is still a resistance — the resistance of the integer gcd — so the relational measurement has not escaped the ln(integer) vocabulary; but it has done something sharper than land on another rung. It has landed on Ω_{gcd}: the subtracted correlation term of the union law of §4,

$$\Omega_{\mathrm{lcm}} = \Omega_a + \Omega_b - \Omega_{\mathrm{gcd}},$$

the one term of the isomorphism's inclusion–exclusion that encodes what a *pair* shares. The mutual information of two modulo channels reaches the resistance isomorphism's Ω_{gcd} term rather than escaping the framework — the relational question, asked in erasure variables, was answered by the lattice core of the isomorphism, the gcd / lcm structure, and not by any single-integer rung *(the identification with Ω_{gcd} is Fact; reading it as the relational core rather than the single-integer ladder is a Reading).* The principle *innovation is connected correlation* thereby acquires its arithmetic face in closed form: the connected correlation between two number-channels *is* $\ln \gcd$. And the coprimality lattice of §4 returns as a correlation pattern — the $6/\pi^2$ density of coprime pairs is now the density of zero-correlation pairs among the channels.

Nor does the correspondence stop at the pair term — a new experiment, run for this paper, shows it is the *whole lattice*. Extending the measurement to banks of three and four channels: the **total correlation** of a bank equals $\sum_i \ln m_i - \ln \mathrm{lcm}$, and this equals the *full alternating inclusion–exclusion over* $\ln \gcd$ *of every subset* — verified for triples and quadruples to 8.9×10^{-16}; the **conditional mutual information** obeys $I(X_1; X_2 \mid X_3) = \ln\left[\gcd(m_1, m_2)/\gcd(m_1, m_2, m_3)\right]$ — conditioning divides out the shared part, the relative meet — to 5.6×10^{-16}; and the **co-information**, the three-way interaction, is exactly $\ln \gcd(m_1, m_2, m_3)$, the triple meet, to 10^{-15} *(Fact; the underlying lcm–gcd identities are elementary).* So the relational core is not one subtracted term of the union law: every multivariate information measure of a channel bank evaluates a gcd-lattice expression, and the entire inclusion–exclusion engine of the divisibility lattice — the structure Part V will identify as the Natural OS — is realised as measured information theory *(the measurements are Fact; "the whole meet-lattice, realised informationally" is a Reading).*

The honest boundary is the §10 circularity at its third door, and it must be drawn exactly. Coprimality *emerges* here as zero correlation, but it is not *derived*: the moduli m_1, m_2 are inputs to the experiment, so what has been built is a correlation-reading of a divisibility relation between given integers — not primes from nothing. The standing problem the gateway named stands. What is genuinely new is the *form* of the test: it is relational, reading the shared prime structure of a pair — something no single-integer resistance can express — and it reproduces the entire coprimality lattice as a correlation pattern. One further direction is flagged and immediately bounded: the corpus's zeta reading is itself an interference story, and whether the continuous critical-line cancellation is the shadow of a discrete proto-phase structure — with the i generated in §12 as the bridge — is a Lead taken up, under a loud no-leverage flag, in §23, and nowhere else.

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15. The Cumulant Foundation: Innovation Is Connected Correlation

The relational result of §14 is an instance of a principle the framework states abstractly — *innovation is connected correlation* — and that principle is not a slogan but a theorem of the statistical mechanics the isomorphism already carries. This section gives the theorem, because Part VI will lean on its central object and §23 on one of its corollaries, and because it supplies the fourth face of §1's correspondence — the gas — in the exact form the erasure reading needs.

The gas is the primon gas: fix $\beta > 1$ and let N be the random integer with $\Pr(N = n) = n^{-\beta}/\zeta(\beta)$. By unique factorisation and the Euler product, $N = \prod_p p^{K_p}$ with the prime occupations K_p *independent*, geometrically distributed with ratio $x_p = p^{-\beta}$ — the Euler product *is* the statement of mode independence — and the energy of a state is its resistance, $\ln N = \sum_p K_p \ln p$, an observable additive over the modes *(Fact, classical; the primon-gas setting is prior art).* Two classical lemmas then do all the work: the m-th cumulant of a geometric variable is $\kappa_m(K) = \sum_{k\geq 1} k^{m-1} x^k$, and for independent variables cumulants add while every *mixed* cumulant vanishes. Composing them gives the master identity, for every order $m \geq 1$:

$$\kappa_m(\ln N) \;=\; \sum_{n\geq 2} \Lambda(n)\,(\ln n)^{m-1}\, n^{-\beta} \;=\; (-1)^m \frac{d^m}{d\beta^m} \ln \zeta(\beta),$$

verified numerically through $m = 4$ against three independent expressions *(Fact, classical).* Here Λ is the von Mangoldt function — $\Lambda(n) = \ln p$ if n is a prime power p^k, and 0 otherwise — making its first appearance in the paper; Part VI is devoted to what it is.

The identity's content, read in the framework's variables, is this: every cumulant of the gas lives on the **innovation skeleton** — atoms exactly at the prime-power resistances, base mass $\Lambda(n)$, weighted by $(\ln n)^{m-1}$ — while composite positions carry *no atom at any cumulant order*. The mechanism is the vanishing of mixed cumulants: a composite is a product of independent modes, and independent things have moments but no connected correlation. The raw moments, by contrast, see everything — $\mathbb{E}[(\ln N)^m]$ carries atoms at every integer. So the passage from moments to cumulants is exactly the passage from all integers to the prime skeleton, and the operator that performs it is the logarithm itself: the moment generating function of the gas is $\mathbb{E}[N^t] = \zeta(\beta - t)/\zeta(\beta)$, and its logarithm is the cumulant generating function. **One logarithm, one more job**: the same map that is resistance, code length, and erasure cost is, probabilistically, the moments-to-cumulants passage — *quantity lives in the moments; the innovation skeleton is what survives the cumulant functor*. This is the statistical identity of the programme's ambition to study the integers stripped of their quantity property, stated as mathematics *(the identity is Fact; the innovation-skeleton reading and the functor framing are Readings)*.

Two corollaries are carried forward. The first is the **order–position lock**: the m-th order statistical form carries exactly $m-1$ extra factors of position, so in a static ensemble the order of a form and the power of its position weight are locked together — the Fisher information of the gas, an $m = 2$ object, carries masses $\Lambda(n) \ln n$, never bare $\Lambda(n)$. The lock is what keeps §23 honest: a functional carrying $m = 1$ masses inside an $m = 2$ shape is *not any cumulant of the static gas*, so whatever its source, it is not this ensemble — an exclusion the critical-line section will need *(Fact, as the m-indexed pattern of the master identity)*. The second is a statement about probes: only mode-additive, composition-respecting observables have cumulants on the skeleton; any function of the scalar $\ln N$ alone re-mixes the modes and its variance sees every integer. *The innovation structure is visible only to probes that respect composition* — the gas's own version of the discipline the whole cartography has practiced *(Reading)*.

One open question is recorded as a Lead and deliberately not banked, because the temptation to bank it is real. There is a classical combinatorial theory in which cumulants are the Möbius inversion of moments over the *partition* lattice; the framework's Möbius function lives on the *divisibility* lattice. Both are instances of the same incidence-algebra inversion, and in the primon gas the single operation ln performs both passages at once — which is why the identification of this section works. But whether that coincidence of two inversions under one logarithm is forced, or is a fortunate alignment specific to free, independent structures, is open; the non-crossing partition lattice of free probability suggests the alignment is structure-dependent, and marks the sharp place to test. Until tested, "the two Möbius functions are one object" remains a Lead *(explicitly unbanked)*.

Part V — The Reframe

16. The Natural Key: Primes Forced by the Filter, Not the Geometry

The cartography has repeatedly met the same wall — primality readable but not derivable, the divisor structure entering through every indexing — and the reframe of this Part begins by asking a different question of the wall itself: not *how do we extract the primes without assuming them*, but *what, exactly, forces the primes to be there at all?* The answer turns out to be locatable, and it is not where the geometry of the framework would suggest.

Half the answer is that a forced basis genuinely exists. Call a resistance *irreducible* if it cannot be split — $\ln n$ that is never $\ln a + \ln b$ for $a, b > 1$. Computed exhaustively to $N = 60$, the irreducible resistances are exactly the primes, and they additively generate every $\ln n$ to machine precision (8.9×10^{-16} maximal error, the factorisation complete in every case), with the generating set unique — which is unique factorisation, wearing its resistance dress *(Fact)*. In the language of data systems: a **natural key exists**. The resistance space is not an amorphous cloud in which any coordinate frame is as good as another; it possesses one distinguished, irreducible, uniquely-generating basis.

The other half of the answer is the surprise: the geometry does not pick that basis. Apply a random rotation to the configuration of ordinals in resistance space, and every pairwise distance is preserved exactly — the geometry is untouched — while the integrality is destroyed: the ordinals, a non-negative integer lattice in the prime basis, become a cloud of irrationals in the rotated frame, verified directly *(Fact)*. Distances, overlaps, everything metric — and, as §18 will show, everything the erasure trace reconstructs — is rotation-invariant, and therefore leaves the basis free up to rotation. What breaks the rotational freedom is the **discreteness**: the requirement that the values form a non-negative integer lattice closed under addition of resistances. Only in the prime basis is the configuration an integer lattice; and that discreteness is nothing other than *succession* — the substrate's stepwise counting, which admits whole steps and no fractions of one.

So the forcing is located, and it splits cleanly by what is assumed. Geometry alone leaves a *surrogate key* — a basis chosen by convention, rotational gauge; geometry plus discreteness forces the *natural key* — the primes. The prime basis is compulsory, but the compulsion comes from the discreteness of counting and the uniqueness of factorisation — from the filter through which the structure reaches us — and not from anything the geometry privileges *(the rotation-invariance of the geometry and the basis-dependence of integrality are Fact; the surrogate-versus-natural resolution is a Reading)*. §18 will sharpen the account with a result obtained for this paper: given the *ensemble* of entities as well as the geometry, the independence principle suffices to select the same basis — discreteness is sufficient for the forcing, but not necessary.

17. The Genesis System Counts; Primality Lives in the Observer's Frame

The location of the forcing licenses the paper's central conceptual move, and it is stated here in full. Everything the cartography requires of the genesis substrate is succession — the counting of §3 — and nothing it has measured requires the substrate to possess anything more. The hypothesis this Part adopts is therefore the simplest one consistent with the whole map: **the genesis rows are plain ordinals** — $1, 2, 3, \ldots$ in the system's own frame, simple succession, nothing prime about them. What reaches us is never the bare ordinal. It is the ordinal passed through the filter the preceding Parts have assembled piece by piece: **uniqueness** — unique factorisation, the one-basis structure of §16 — and **irreversibility** — the ln of Landauer, which makes resistance additive over products and is the only channel across the barrier. And the shadow that plain counting casts through that filter is the inverse prime. Primes are what ordinals look like from here. **Primality is a property of the observer's frame — it exists here, not there.**

The move must immediately be graded, because its epistemic standing is exactly fixed by Axiom 5 and the paper does not exceed it. The substrate stays behind the barrier; no measurement reaches it; so "the genesis rows are plain ordinals" is not a deduction and cannot be one. It is the *simplest hypothesis consistent with the shadow* — everything in the map is explained by counting plus the filter, and nothing in the map requires more — and it is held as a **Reading**: supported, economical, and interpretive. No claim is made to see behind the barrier *(Reading, with the boundary stated).*

What the move does to the programme's target is the deeper consequence, and it is a demotion that clarifies. The earlier stages of this work spoke of reading the genesis system's *meaning* through its erasures. But if the rows are bare positions, there is nothing to refer: positions do not mean. The meaning-geometry that the reconstruction work recovers (§18) is real, but it is the structure of the *shadow* — the multiplicative organisation that uniqueness and irreversibility stamp onto succession — not a semantics the system holds. The honest target of the whole programme is therefore not a semantics but a **data structure**: behind the prime and multiplicative shadow, the substrate is the free monoid on succession — counting, and only counting — and the entire prime lattice, the gcd / lcm joins, the divisibility order, is that monoid's unique-factorisation shadow. We read how counting, filtered, *organises* — not what it *means*. This is the reframe the whole paper turns on: erasure reveals the genesis system's data structure, not its meaning.

18. The Natural OS Recovered Backward, and the Syntax/Semantics Filter

The reframe makes a prediction of sorts about the corpus itself, and the prediction checks. If the shadow of filtered counting is the divisibility lattice, then the data structure the erasure trace recovers should be exactly the structure the programme's earlier work — the Natural Operating System of the companion Paper 2 — built by the opposite route. Paper 2 constructed the Natural OS *forward*: starting from the integers' own organisation, it assembled the divisibility lattice with its Möbius function as the master inclusion–exclusion engine of the number system. The present work arrives at the same lattice *backward* — from the erasure trace, with succession underneath and unique factorisation as the filter that turns counting into the lattice. Same operating system, opposite direction of construction; the convergence of the two routes on one structure is itself evidence that the structure is the invariant thing *(Reading — the connection).*

That the backward route genuinely delivers the structure is not asserted but demonstrated, in a working model built to test exactly this. Fix a hidden semantics — primitive meanings assigned to the primes $\{2, 3, 5, 7, 11\}$ with resistances $\ln p$, composite meanings their squarefree products — and expose *only* its erasures: the meet-erasure of relating two meanings (keep the shared part, the gcd; forget the symmetric difference; observe the resistance forgotten) and the reset-erasure of forgetting a meaning outright. From that trace alone, four things reconstruct. The meet-erasure is *exactly* the squared distance in meaning-space, $E(a, b) = \|u_a - u_b\|^2$ under the natural feature embedding — the erasures *are* the meaning-geometry, delivered as pairwise distances *(Fact).* The shared meaning recovers to machine precision, $\Omega(\gcd(a, b)) = \frac{1}{2}(\mathrm{norm}_a + \mathrm{norm}_b - E(a, b))$, exact to 6.7×10^{-16} *(Fact).* The number of primitive meanings recovers exactly: classical multidimensional scaling on the meet-erasures returns embedding dimension five — the true count of hidden primes — with eigenvalues $[8.49, 5.00, 2.77, 1.85, 0.31]$ and then zero *(Fact).* And the full meaning-geometry recovers up to isometry: after the single best rotation, the reconstructed meaning-vectors match the true ones to 4.8×10^{-15} *(Fact).*

Up to isometry — and there is the filter. The geometry crosses the barrier intact; the *labels* do not. Which axis is which meaning — the assignment of meanings to specific primes — is precisely the rotational gauge that §16 showed the geometry cannot see, and it is not in the trace *in principle*: rotating the configuration changes no distance and therefore no erasure. This is the **syntax/semantics filter**, confirmed in a working model: erasure passes the meaning-*structure* — the geometry, the overlap lattice, the primitive count — and withholds the *syntax* — the labels, which remain surrogate keys, free *(the reconstruction facts are Fact; the filter reading is a Reading,* *strongly grounded).* In the vocabulary of §16 the two results are one: the trace recovers everything rotation-invariant, and the natural key is forced only when the discreteness is added — which is why the recovered structure is the lattice and not the labelling.

The clean case declares its own limits, and they are recorded as the frontier rather than smoothed over. The test was given exact erasures, every pair, with each event's type known; real reconstruction faces noise — under which the faintest primitives wash out first, the 0.31 eigenvalue before the 8.49 — missing pairs, and undifferentiated events, where sorting "relate" from "forget" is itself an inference. And the sharp question the test crystallised — MDS returns the room up to rotation, with the true prime axes one special set inside it; can those axes be singled out from the geometry by a structural principle? — has now been *run*, in a new experiment for this paper, with two candidate principles and opposite outcomes. **Sparsity fails**: the varimax rotation, the classical sparsity-seeking criterion, does not find the prime axes (worst axis alignment $|\cos| = 0.488$) — the informative negative. **Independence succeeds exactly**: independent component analysis recovers all five prime axes at $|\cos| = 1.000000$, returns the axis scales equal to $\sqrt{\ln p}$ to 3.6×10^{-15} — the resistances read back off the geometry — and degrades gracefully under noise (worst alignment 0.986 at feature-noise 0.20) *(Fact).* And the principle that succeeds is the framework's own: under the uniform entity ensemble the true prime coordinates are statistically *independent*, so the prime basis is the unique **independence basis** of the meaning space — the geometric face of the same independence that is coprimality in §14 and the Euler product's mode independence in §15, its third appearance in the paper. The axis *labels* remain gauge — the components return unordered, named only by their recovered resistances — so the syntax/semantics filter holds exactly where this section drew it. The §16 conclusion is sharpened rather than overturned: distances alone still cannot fix the basis; what fixes it is the geometry *plus the ensemble*, with independence as the criterion — discreteness was sufficient but is not necessary. The general case — non-uniform ensembles, partial observation — remains open *(the recovery results are Fact; the independence-key reading is a Reading; the general case is a Lead).*

19. Bearing on Primality: A Computational Account, Not a Quantitative Derivation

What the reframe does to the standing problem of primality must be stated with precision, because it does two different things and the paper's honesty requires keeping them apart.

The first is a genuine contribution: the reframe gives primality a **computational identity**. A prime is an *irreducible resistance* — an ordinal whose forgetting-cost is not a sum of smaller forgetting-costs — a structural property of the shadow, not something the genesis system computes or contains. Under this account the multiplicative structure of arithmetic is *intrinsic to the shadow*, handed over by the filter itself — the ln from irreversibility, the integer lattice from succession, the uniqueness from factorisation — rather than imported into the analysis from outside. That is an answer, in the framework's terms, to the question *what is a prime?*: it is the smallest indivisible act of forgetting, the atom of the erasure ledger, the point at which the shadow's additive structure bottoms out. The account explains where primes come from — they are what the filter makes of counting — and why they are compulsory — the natural key of §16 *(Reading; a genuine contribution of the reframe).*

The second thing must be said just as plainly: this account is **not linked to the quantitative content of primality**. *Which* integers are prime — the location of the irreducibles along the ladder, the distribution that number theory studies — is still established only by engaging the divisor structure, because testing whether a resistance is irreducible is testing whether its integer has a proper divisor. The reframe therefore **relocates** the circularity that §10, §13, and §14 met at three doors — moving it out of the experimental indexing and into the filter itself, where it now stands as a property of the shadow rather than an artifact of our probes — but it does not remove it. Whether the prime-signal can ever be reached without the multiplicative structure it is defined against remains the standing open problem of the programme, unresolved here as everywhere *(the residual circularity is flagged; the differentiation between the computational account and the quantitative content is load-bearing and recurs at §25).*

The gain, then, stated exactly: the paper explains primality *computationally* — what a prime is, and why the shadow must contain primes — without deriving it *quantitatively* — which numbers are prime, free of the divisor lattice. The next Part closes the arithmetic arc by exhibiting the primality signal itself in closed form, and the closed form will make the relocated circularity concrete: the signal is an inclusion–exclusion over exactly the lattice the account cannot do without.

Part VI — The Primality Shadow

20. The Primality Shadow Is the von Mangoldt Function

Part V established what a prime is in the framework's terms — the irreducible survivor of the filter. This Part exhibits the survivor's signal in closed form. The originating proposal, stated in the data-structure vocabulary of §17, was that the primality shadow of a number is a signed ledger: the ON state, minus the erasure of everything that is not the unique row, plus the nats that certify irreducibility — the irreversibility test, written as an accounting. Formalised, that ledger is a classical object, and the formalisation is exact.

Two identities carry it, both verified to 10^{-12} across the range tested. The first builds resistance from innovations:

$$\ln n = \sum_{d|n} \Lambda(d),$$

every forgetting-cost the sum of the von Mangoldt masses of its divisors — the resistance ladder assembled from the innovation skeleton of §15, rung by rung. The second inverts it, and is the irreversibility test itself:

$$\Lambda(n) = -\sum_{d|n} \mu(d)\, \ln d,$$

a signed inclusion–exclusion over the divisor lattice — the Möbius engine of the Natural OS, §18's recovered structure, here running as the extractor — that pulls the prime-signal back out of the resistance, with $\Lambda(n) = \ln p$ if $n = p^k$ and 0 otherwise *(both identities classical; Fact).*

Read as the proposed ledger, the inversion's three terms map exactly, and the mapping is cleanest where it matters. For a prime — take $n = 5$ — the ledger has two rows only: the divisor $d = 1$ is the **ON state**, $\mu(1) = +1$ contributing $\ln 1 = 0$, the identity always present; the divisor $d = 5$ is the **certificate**, contributing $+\ln 5$; and there is no composite divisor, hence **no erasure term** — the full $\ln 5$ survives, $\Lambda(5) = \ln 5$. The prime *locks up*: there is nothing to reduce it. For a composite — $n = 6$ — the erasure appears: the certificates $+\ln 2 + \ln 3$ enter for the prime parts, the term $-\ln 6$ cancels them exactly, and $\Lambda(6) = 0$. The composite *reduces away.* The verification exhibits the cancellation row by row — $[+0.693] + [+1.099] + [-1.792] = 0$ — and the survival — Λ equal to $\ln p$ at every prime power in the range, zero at every composite *(Fact)*. The erasure column of the ledger is the inclusion–exclusion cancellation that annihilates everything not irreducible, and **a prime is exactly a number with an empty erasure column** — the row with nothing to cancel *(recognising the classical identity as the data-structure ledger is a Reading)*.

A new experiment, run for this paper, then lifts the ledger from a combinatorial identity to a *measured* quantity, and in doing so fuses this Part with Part IV. Measure, empirically, the conditional entropy of the channel $a \bmod n$ given *all of its divisor channels* — everything the sub-channels of n already report. The result, verified for $n = 2..36$ to 6.7×10^{-16}, is

$$H\big(a \bmod n \,\big|\, \{a \bmod d : d \mid n,\ d < n\}\big) \;=\; \Lambda(n):$$

a prime power *innovates* — carries information beyond its divisor channels — by exactly $\ln p$, and a composite carries exactly nothing new, fully redundant given its parts. **The von Mangoldt function is the channel's innovation**, in precisely the sense of *innovation is connected correlation*; and the building identity $\ln n = \sum_{d|n} \Lambda(d)$ becomes a chain rule — the channel's information is the accumulated innovation of its divisor tower, verified directly ($n = 12, 16, 24, 30, 36$, to 4.4×10^{-16}) *(the measurements are Fact; the underlying lattice identity — the lcm of the proper divisors of n is n/p for a prime power and n otherwise — is elementary; the innovation reading is the contribution)*. The atoms of §15's innovation skeleton, whose masses $\Lambda(n)$ entered there through the gas, are thereby *measured* at the channels: the primality shadow is not only the Möbius inversion of the resistance — it is the conditional entropy of counting read modulo n, given every coarser reading. A prime is a channel that says something new.

And a further experiment, prompted by that formulation, advances the standing circularity itself by one genuine step. The conditioning set above was the *divisor* channels of n — still the lattice at the door. Replace it with a set defined purely by **succession order** — *all predecessors*, every channel $2..n{-}1$ — and the signal survives intact:

$$H\big(a \bmod n \,\big|\, \{a \bmod k : 2 \le k \le n-1\}\big) \;=\; \Lambda(n),$$

verified by exact enumeration for $n = 2..18$ to $6.2{\times}10^{-15}$, with divisor knowledge appearing *nowhere in the extraction*: the procedure is the ordinal sweep of the Stage-1 channel bank plus a conditional entropy, and nothing else *(Fact; the underlying identity —* $\ln \operatorname{lcm}(1..n)$ *is the second Chebyshev function* $\psi(n)$*, with* $\Lambda(n) = \psi(n) - \psi(n-1)$ *— is classical, flagged)*. Three corollaries sharpen it. The novelty ratio innovation$/\ln n$ equals 1 *exactly when n is prime* — a prime is a channel whose innovation over everything before it is its entire content; a prime power innovates by the fraction $1/k$; a composite by zero *(Fact)*. The cumulative innovation of the sweep is $\psi(n)$ itself — the total information of the ordinal bank is the central object of prime distribution, and the prime number theorem $\psi(n) \sim n$ then reads, in the framework's variables, that *counting, read through all its cycles, produces new information at asymptotically one nat per step*; the fluctuations of that rate are the domain of the deepest open questions of prime distribution, on which no leverage is claimed or implied *(the PNT is classical; the information-rate reading is a Reading; nothing further is claimed)*. And the sweep-profile is the factorisation read by order: a prime's conditional entropy stays flat at $\ln n$ through the entire sweep, while a composite's steps down exactly as the sweep covers each prime-power part *(Fact)*. The circularity is thereby weakened with precision: the divisor lattice is no longer an input to the *procedure* — the extraction is succession-native, and primality emerges as its output — while the relabel caution stands at the family level (the channels are still the integers' cycles) and the divisor structure has not vanished but moved *into the data*, living now in the joint distribution of the readouts rather than in the method. The relocation of

§19 advances from "the extraction needs the lattice" to "the extraction needs only order; the lattice announces itself in what the measurements say."

The honest boundary is the one §19 predicted, now concrete. The extraction of $\Lambda(n)$ runs over the divisors of n: the ledger *is* an inclusion–exclusion over the divisor lattice, so obtaining the prime-signal still requires the multiplicative structure the signal is defined against. The closed form deepens the account — the circularity now has an exact shape, a specific lattice, a named inversion — but it does not dissolve the loop; the standing problem of §19 stands *(flagged; consistent with "relocated, not removed")*.

21. Three Faces of Unique Factorisation

One identity remains, and it closes the arithmetic arc by joining the paper's three main arithmetic results into a single object. The von Mangoldt series is the logarithmic derivative of the Euler product:

$$-\frac{\zeta'}{\zeta}(s) = \sum_n \frac{\Lambda(n)}{n^s}, \qquad \zeta(s) = \prod_p \left(1 - p^{-s}\right)^{-1}$$

(Fact, classical). So Λ — the primality shadow of §20 — is precisely the prime-signal the Euler product hands back when differentiated: the bridge between the additive sum over all integers and the multiplicative product over the primes, written in resistance variables.

And the product factorises for a reason the cartography has already measured. The Euler product is the statement that divisibility by distinct primes is *independent* — the mode-independence on which the whole gas of §15 stood — and that independence is exactly the coprimality result of §14: $I(a \bmod p\,;\, a \bmod q) = \ln\gcd(p,q) = 0$ for distinct primes. The factorisation of the product and the vanishing of the correlation are one fact in two vocabularies.

Three things are therefore one. **The Euler product** — the multiplicative organisation of the integers, the forward construction of the Natural OS; **the** $\ln\gcd$ **coprimality** — the relational measurement of Part IV, prime-sharing as correlation; and **the primality-shadow ledger** — the von Mangoldt inclusion–exclusion of §20, primes as the irreducible survivors: three faces of unique factorisation written in information. The product factorises because the prime channels are uncorrelated; its log-derivative is the von Mangoldt; and the von Mangoldt is the ON-minus-erasure-plus-certificate ledger with primes the rows that survive. Each face was reached by a different route in this paper — the product through the gas, the correlation through the channels, the ledger through the data-structure reframe — and they meet because they were always the same theorem, unique factorisation, seen from the statistical, the relational, and the combinatorial side *(the identities are Fact; the synthesis across the three faces is a Reading)*.

The honesty note for the Part is short and total: **no new mathematics is claimed anywhere in it.** The von Mangoldt, Möbius, and Euler identities are textbook analytic number theory. The contribution is the recognition that the data-structure primality-shadow proposal lands exactly on them, and the synthesis with the erasure cartography and the coprimality result — the demonstration that the framework's independently-motivated objects converge on the classical core rather than beside it. The value is coherence, not a theorem; and the coherence is the point, because a hypothesis that claims number theory as a shadow is tested precisely by whether its native constructions land on number theory's own load-bearing identities. They do.

Part VII — Frontier and Honest Boundary

22. The Genesis Bit: One Phenomenon, Five Faces

A single point has recurred through every Part of this paper, and before the frontier is drawn it should be collected, because the recurrence is itself a finding. The point is $g = 2$, $p = 2$, $n = 1$ — the genesis bit, the first step from the boundary into the interior — and it has appeared five times, in five vocabularies, always as the distinguished case.

In the algebra (§5), $p = 2$ is where the Weyl braiding closes on the Pauli algebra and the imaginary unit is realised — $i = \sqrt{\mathrm{NOT}}$, the square root of the substrate's one primitive — and simultaneously where the probe is at once maximally informative (a full bit) and maximally disturbing (a full π), the sharpest point of the information–disturbance complementarity. In the channel thermodynamics (§10), $g = 2$ is the unique lossless channel — the one girth at which the destructive probe erases nothing, the anchor from which extraction efficiency falls — and, in the stream, the cheapest source of information the system can emit. In the merge (§11), $n = 1$ is where the two-preimage scattering space carries exactly the Pauli algebra, and (§12) where the commutator of succession and the probe *is* the phase generator, $[S, D] = -i\sigma_y$. And in the critical-line bouquet (§23, immediately below), the alternating sign that tames the divergent sum into a convergent one is the nontrivial character mod 2 — the 2-phase — while the simplest exact cancellation in the discrete seed is the genesis bit's own $1 + (-1) = 0$.

Each face is a Fact where it was established; the finding of this section is the Reading that unifies them: these are not five coincidences but **one phenomenon seen five times** — the first interior step is simultaneously the birthplace of phase, the lossless channel, the cheapest source, the Pauli case of the merge, and the convergence-maker of the bouquet, because in every vocabulary it is the same structural object: the minimal distinction, the two-element case, where the alternative is unique and nothing remains to coarsen. Wherever the framework's machinery degenerates to its cleanest case, it degenerates to the same place, and that place is the substrate's first act *(Reading)*.

23. The Critical-Line Bridge: Mechanism, No Leverage

This section is RH-adjacent, so its boundary is stated before its content, and the boundary governs everything in it: **this section claims nothing about the Riemann Hypothesis and gives no leverage on the location of the zeros.** The zeros are classical; their values are taken as known throughout; and the bouquet studied below must cancel at them, because it *is* the analytically-continued zeta function. What the section contains is a mechanism made literal and two identifications — nothing more, and it should not be read as more.

The discrete seed is already in the substrate's hands. The proto-phase of §12 contains total destructive interference natively: the d-th roots of unity sum to zero, to machine precision, for every d — at $d = 2$ it is the genesis bit's own $1 + (-1) = 0$, the NOT cancellation — uniform amplitudes and uniform phases cancelling exactly *(Fact)*. The continuous object is the resistance-phasor bouquet on the critical line,

$$\eta\left(\tfrac{1}{2} + it\right) = \sum_{n \geq 1} (-1)^{n-1}\, n^{-1/2}\, e^{-it \ln n},$$

the same idea with log-spaced phases $t \ln n$ and amplitudes $n^{-1/2}$. Computed directly, the bouquet's magnitude falls to $\sim 10^{-7}$ exactly at the known zeros — 2.7×10^{-7} at $t = 14.1347$, 8.4×10^{-7} at 21.0220, 9.7×10^{-7} at 25.0109, 3.6×10^{-7} at 30.4249 — while running $O(1)$ at the midpoints between them (1.42, 3.49, 1.70), a coarse scan placing the interference nulls at the zeros and nowhere else

(Fact). The critical-line cancellation is therefore *literally* destructive interference of phasors — the corpus's zeta picture, previously an interpretation, now a confirmed computation *(the numerics are Fact; "the mechanism is phasor interference" is a grounded Reading)*. One deformation is explicitly *not* available and is disclaimed: the uniform roots-of-unity cancellation does not deform into the zeta cancellation. They are the same mechanism — interference nulls — one trivial and uniform, one hard and logarithmic, and no derivation of the second from the first is claimed or attempted.

What the bridge genuinely gives is connective tissue, in two identifications. First, **the spin rates are the forgetting-costs.** The phases in the bouquet are $t \ln n$, and $\ln n$ is the resistance — the Landauer erasure cost that the entire cartography of Part III catalogued, operation by operation. The integers in the zeta cancellation spin at rates equal to their erasure costs; the nontrivial zeros are interference nulls of integers turning at the rates it costs to forget them *(that* $\ln n$ *is both quantities is Fact; the identification is the content, a Reading)*. Second, **the genesis bit bookends the picture.** The bouquet converges on the critical line only because of the alternating sign $(-1)^{n-1}$ — the nontrivial character mod 2, the 2-phase, the eta/zeta factor being literally $(1-2^{1-s})$ — so the same object that supplies the first and simplest discrete cancellation also tames the divergent integer bouquet into a convergent one: the seed of cancellation at one end, the boundary of convergence at the other, the fifth face of §22 *(the alternation as the character mod 2 is Fact; the bookend framing is a Reading)*. One Lead is flagged and left closed: whether other prime characters — the Dirichlet L-function layer — play roles analogous to the 2-phase is a natural question, and it is not pursued. The order–position lock of §15 stands guard over this whole territory: the Weil-type functionals of the corpus's zeta reading carry $m = 1$ masses in $m = 2$ shapes and are therefore not cumulants of the static gas, so nothing in the static ensemble of this paper reaches them — an exclusion, not an opening. **No leverage on the zeros is claimed, and this section should not be pushed to pretend otherwise.**

24. Parked Threads

The programme holds several directions deliberately open, and the discipline of the paper is to record them as flagged, parked threads — each with the reason it is parked — rather than to pursue any of them past its warrant.

The physical kill-or-promote. The one genuine falsifiability path for the erasure axiom is physical heat measurement — a test that could actually come out wrong, which §8 established no simulation can be. The natural vehicle is the primon gas of §15 realised as thermodynamics: a system with mode energies $\ln p$, partition function $\zeta(\beta)$, where the framework's entire stack — resistance, erasure, the innovation skeleton — becomes measurable physics. This is the promotion path for the paper's central Lead, and it is the most important parked thread. A methodological rule travels with it: physics is to be used as *confirmation*, never as *source* — the framework's claims must be derived in its own terms and then tested against physical measurement, not read off physics and then decorated.

The logarithmic filter as a physical shadow. The logarithm is the universal signature of scale-invariance — the additive readout of multiplicative structure — and the same ln that this paper reads as erasure cost appears wherever a physical or perceptual system encodes scale-free input efficiently. Whether the resistance isomorphism belongs to that family as one bridge among several, with the primon gas as its cleanest entry point, is a framing question for the physical direction, parked with it *(Lead)*.

Prime distribution: the flow and the echo. The corpus's verified flow test — the Riemann–

Weil explicit formula, confirmed numerically to 10^{-24} in earlier work — reads the primes as atoms and the zeros as echo-frequencies of a flow. The new angle the present work adds is a gauge observation: *relative ordinality* — counting with no absolute origin — is the translation freedom of the resistance line, and translation is Fourier-dual to phase, linking the flow picture to the phase-from-the-probe construction of §5 and §12. The thread is RH-bounded in exactly the sense of §23 — picture-deepening, not leverage — and is parked as such *(Lead)*.

The cross-modulus divisibility pattern. The gateway of §13 located the multiplicative and prime structure in the pattern of divisibility across all moduli — the channel-bank — and §14 gave the pattern its first relational measurement. The ordinal-innovation result of §20 has now advanced this thread one genuine step: the prime signal is extractable by an order-defined sweep of the bank, with the divisor lattice no longer an input to the procedure. What remains open is the relabel at the family level — the channels are still the integers' cycles — and whether the family itself can be made to *earn* its indexing through the ledger, the generative inversion the programme has held as its hardest target from the start *(Lead; the standing open problem, advanced but not closed)*.

The independence key beyond the clean case. §18's recovery question was answered for the clean, uniform-ensemble case in this paper — the natural key is recoverable by the independence principle, with sparsity's failure the honest negative — but the general case is open: non-uniform ensembles, partial and unlabelled observation, and whether the independence criterion continues to select the prime axes when the entity distribution is not the free product the clean case assumed *(Lead)*.

The observation-probability lead. The stream economics of §10 — information cheapest to generate at the genesis bit — suggests a connection between erasure cost and observation probability: *cheaper to generate, more probable to observe*. Deferred in the originating work and deferred here *(Lead, unexplored)*.

The two Möbius functions. §15's open question — whether the coincidence of the partition-lattice and divisibility-lattice inversions under one logarithm is forced or is an accident of free structure, with free probability's non-crossing lattice as the sharp test — is parked, explicitly unbanked *(Lead)*.

The merge dynamics and arithmetic phase. §12 established the phase as structurally available on the merge's configuration space; whether the merge *dynamics* produce a definite phased state, whether interference then fires, and whether the additive system can be coupled to multiplicative structure so that phase becomes arithmetic rather than positional — these are the doors §12 named and did not open *(Lead)*.

Part VIII — Discussion

25. What Is Established, and What Is Not

The paper closes by stating its reach plainly, in both directions.

What is established is a coherent, internally-consistent account of number theory as the erasure trace of a quantity-free counting substrate, and the account is not a single claim but an assembled structure, each joint of which has been verified in its own right. The quantity-free floor stands: from the single primitive of distinction, the substrate counts (succession dissolving the finite-symmetry obstruction), remembers (the integer lattice), computes universally (FRACTRAN), projects onto the mature resistance structure (the isomorphism, unique up to the scale that Landauer fills), and manufactures its own phase (the Weyl braiding, closing on $i = \sqrt{\mathrm{NOT}}$ at the genesis bit) — with quantity nowhere assumed and everywhere paid for. The erasure turn stands on its one hard pillar — resistance *is* the Landauer cost — and its accounting is confirmed end to end: energy located only at erasures, equal to the resistance erased, reversible steps free, both kill conditions untriggered, the books balanced on every ensemble. The cartography stands: every irreversible operation forgets the conditional entropy of its input — ln(fan-in) under maximum ignorance — and the two-legged ledger is universal across every substrate tried, additive, rewriting, and modular, with the multiplicative structure entering exactly at modulo, where the ledger becomes a factor pair. The relational core stands: the mutual information of two residue channels is $\ln \gcd$, coprimality is exactly zero correlation, the result lands on the Ω_{gcd} term of the union law — the framework closing on itself rather than escaping — and the whole inclusion–exclusion of the gcd lattice, not one term of it, is realised in the multivariate information measures of channel banks. The reframe stands as the account's interpretive spine: the prime basis is a natural key forced by discreteness and uniqueness, not by the rotation-free geometry — and now shown recoverable from the geometry by the independence principle, coprimality's own criterion, where sparsity fails; the substrate, on the simplest hypothesis consistent with the whole map, merely counts; primality lives in the observer's frame; and the recovered data structure is the divisibility lattice of the Natural OS, obtained backward from its erasure trace, with the syntax/semantics filter demonstrated in a working model and the phase-carries-the-split claim upgraded to an operational protocol. And the arithmetic arc closes: the primality signal is the von Mangoldt function via Möbius inversion — and directly, by measurement, the conditional entropy of a channel given its predecessors, extractable by succession order alone, with a prime the channel of pure novelty, cumulative innovation the second Chebyshev function, and the Euler product, the $\ln \gcd$ coprimality, and the primality ledger three faces of unique factorisation written in information.

What is not established is stated with the same care, because the paper's claim to be believed rests on it. The genesis system's existence is not established and cannot be: it is unfalsifiable by construction, the barrier being the complement of every probe we hold, and the erasure axiom is accordingly held throughout as a corroborated *consistent accounting* — its erasures correctly located, its books balanced — and not as a physical fact; the one genuine falsifiability path, physical heat measurement in the primon-gas direction, is parked and open. The quantitative content of primality is not derived: the paper gives primality a computational identity — an irreducible resistance, the atom of forgetting, forced by the filter — but *which* integers are prime is still established only through the divisor lattice, the circularity relocated into the filter and made exact in the von Mangoldt inversion, not removed. And nothing about the Riemann Hypothesis is established, approached, or implied: the critical-line material confirms a mechanism — phasor interference, with spin rates equal to forgetting-costs — at zeros whose locations were taken as known, and the order–position lock actively excludes the static ensemble of this paper from the functionals that bear on the zeros' location.

The contribution, restated in one sentence for each of its parts: the axioms and the genesis framing; the erasure axiom with its pillar and its honestly-bounded corroboration; the systematic cartography of number theory in erasure variables, organised by the fan-in law and the universal two-legged ledger; the relational result and its identification with the union law's correlation term; the reframe of the substrate as counting, with the natural-key forcing located and the syntax/semantics filter demonstrated; the computational account of primality, differentiated from its underived quantitative content; the synthesis of the three faces of unique factorisation; and a small number of new formal observations, each marked where made. No new theorem is claimed anywhere. The discipline that governed the work governs its close: the paper refuses to claim a breakthrough where only an established fact has been found — and holds that the establishment of coherence, on this scale, across these domains, graded claim by claim, is itself the result.

Appendix A — Verification Map

Every Fact in the paper that rests on computation is listed here with its verifying script and the confirmed figure, as re-run for this draft. All scripts are in the project corpus; all figures were reproduced in the session that produced this paper.

Result (section)	Script	Confirmation
Chain rule $\ln g = H(1/g)+(1-\frac{1}{g})\ln(g-1)$, every girth; reversible tick = 0; overwrite = $\ln g$ (§7, §10)	`stage1_erasure_channels.py`	identity exact for $g = 2..12$; g=6: $0.45056 + 1.34120 = 1.79176$ both routes
Genesis bit lossless; efficiency decline (§10)	`stage1_erasure_channels.py`	g=2: erased = 0.000000; efficiencies $1.000, 0.579, 0.311, 0.211, 0.127$
Sub-period test returns the primes (relabel) (§10)	`stage1_erasure_channels.py`	primes 2..47 exactly, no factoring
Stream: coarsen + reset = $\ln L$; additivity iff coprime (§10)	`stage1_stream.py`	$(2, 3)$: $0.462 + 1.330 = 1.792$; redundancy 0 iff coprime; $(2, 4)$: 0.216; $(2, 4, 8)$: 0.419
Merge-erasure $\ln(m + 1)$; temperature-independence; two-legged ledger (§11)	`stage1_additive_merge.py`	ledger = 4.3944 (uniform), 5.0040 (geometric), exact
Build-erasure a path function; floor $\ln(N + 1)$; smallest-first minimises (§11)	`stage1_build_paths.py`	N=8: 2.1972 (single) to 12.1087 (linear); balanced 9.8105
Reverse-scattering entropy = $\ln(n + 1)$; Weyl relation; n=1 → Pauli/$\sqrt{\text{NOT}}$; interference beyond mixtures (§11)	`reverse_merge_weyl.py`	equality and $ZX = \omega XZ$ for $n = 1..5$; $\sqrt{\text{NOT}}^2 = \text{NOT}$; amplitude cancellation exhibited
Z generated from $\{S, D\}$; $[S, D] = -i\sigma_y$ at n=1 (§12)	`phase_from_merge.py`	both true for $n = 1..5$; commutator identity exact
Phase cost = merge-erasure at every n (§12)	`phase_cost.py`	$0.6931, 1.0986, 1.3863, 1.6094, 2.1972$ at $n = 1, 2, 3, 4, 8$, exact
Per-split angle positional, per-split cost uniform (§12)	`phase_cost.py`	n=8: 40° steps; cost $\ln 9$ every split
Additive catalogue: flat/graded fan-ins; ledgers close (§13)	`erasure_operations.py`	totals 4.3944 / 2.1972 exact; max fan-in $2m$+1, min $2(M-m)$+1

Result (section)	Script	Confirmation
Rewriting: conditional erasure; ranking; totals = $L \ln 2$ (§13)	`rewriting_erasure.py`	19/32 erased; means $0.7798 > 0.5689 > 0.4496$; totals 3.4657
Modulo: quotient fan-in; factor-pair ledger (§13)	`modulo_erasure.py`	$M{+}1{=}12$: fan-ins $12, 6, 4, 3, 2, 1$; $\ln \frac{M+1}{m} + \ln m = \ln 12$
$I = \ln \gcd$; coprimality = zero correlation (§14)	`channel_correlation.py`	max deviation 2.2×10^{-16} over all pairs 2..8
Cumulant master identity (§15)	(mpmath, in corpus note)	verified $m = 1..4$ at $\beta = 2$ via three independent expressions, 12 digits
Irreducible resistances = primes; generation; rotation vs discreteness (§16)	`natural_key_test.py`	primes to 60 exact; max generation error 8.9×10^{-16}; distances rotation-invariant, integrality not
Meet-erasure = squared distance; $\Omega(\gcd)$ recovered; MDS dimension = 5; geometry to isometry (§18)	`toy_language_reconstruction.py`	errors 6.7×10^{-16} and 4.8×10^{-15}; eigenvalues $[8.49, 5.00, 2.77, 1.85, 0.31]$
$\ln n = \sum_{d\mid n} \Lambda(d)$; $\Lambda = -\sum \mu(d) \ln d$; survive/cancel (§20)	`primality_shadow_ledger.py`	both identities for $n = 2..12$; $\Lambda(6)$ cancellation row-by-row
Composed energy ledger; both kill conditions untriggered (§8)	`energy_ledger.py`	erasure $= 1.7918 = \ln 6$ (uniform), $H(b \mid a) = 1.6587$ (random); non-erasure max 4.4×10^{-16}; discrepancy $\leq 2.2 \times 10^{-16}$
Roots of unity cancel; bouquet nulls at the classical zeros (§23)	`weyl_clock_cancellation.py`	$\leq 9.2 \times 10^{-16}$ for $d = 2..6$; $\sim 10^{-7}$ at $t = 14.135, 21.022, 25.011, 30.425$; $O(1)$ between
Multivariate lattice information: total correlation = full gcd inclusion–exclusion; conditional MI = relative meet; co-information $= \ln \gcd(\text{triple})$ (§14)	`lattice_information.py`	8.9×10^{-16}, 5.6×10^{-16}, 10^{-15} respectively, triples and quadruples
$\Lambda(n) = H(a \bmod n \mid$ divisor channels); $\ln n =$ accumulated innovation of the divisor tower (§20)	`channel_innovation_vonmangoldt.py`	6.7×10^{-16} over $n = 2..36$; tower chain rule to 4.4×10^{-16}
Split recovery from the coherent phase = 1; after decoherence = $1/d$ exactly (§12)	`phase_recovery.py`	exact at $n = 1, 2, 4, 8, 12$

Result (section)	Script	Confirmation
Natural key by independence: ICA recovers all prime axes ($\lvert\cos\rvert = 1.000000$), scales $= \sqrt{\ln p}$; varimax fails (0.488) (§18)	`sparse_axis_recovery.py`	scale recovery to 3.6×10^{-15}; noise-robust to 0.986 at sd 0.20
Ordinal innovation: $H(a \bmod n \mid$ all predecessors$) = \Lambda(n)$; novelty ratio = 1 iff prime; cumulative = $\psi(n)$; decay profile = factorisation (§20)	`ordinal_innovation.py`	6.2×10^{-15} over $n = 2..18$; profiles exact

Appendix B — The Classical-Results Honesty Boundary

The following classical results are used throughout and are not contributions of this work; each is flagged in the text where it carries load. Landauer's principle and Bennett's reversible-computation results (erasure cost, the demon's reset, dissipation-free reversible steps). The Shannon quantities: entropy, conditional entropy, the chain rule, mutual information. Jaynes's maximum-entropy principle (the superposition state). The Weyl clock-shift algebra and the Pauli algebra. The von Neumann entropy of decoherence. Cauchy's functional equation (uniqueness of the logarithm up to scale). Unique factorisation; the linear independence of $\{\ln p\}$ over the rationals; the union law $\mathrm{lcm} \cdot \gcd = ab$; the coprimality density $6/\pi^2 = 1/\zeta(2)$. The Chinese Remainder Theorem (independence of coprime channels). Spencer-Brown's mark; the anharmonic group of the cross-ratio. Conway's FRACTRAN and Minsky's register machines (universality). The Möbius function, the von Mangoldt function, Möbius inversion, and the Euler product with its logarithmic derivative; Rota's incidence-algebra inversion (both the divisibility-lattice and partition-lattice instances). The primon-gas construction (prior art). The geometric distribution's cumulants; the additivity and mixed-vanishing of cumulants for independent variables. Classical multidimensional scaling and Procrustes alignment. The symmetric-difference metric on weighted sets. Optimal-merge (Huffman) ordering. The Dirichlet eta function, its relation to zeta, conditional convergence on the critical line, and the locations of the nontrivial zeros used in §23. The contribution of this paper is interpretive and synthetic — the framing, the cartography, the reframe, and the marked formal observations — and at no point a new theorem.

Appendix C — The Proto-Glossary

The discipline: a proto-term is retained only while it carries both a definite pre-quantitative meaning and a named quantitative completion; a proto-object with no completion, or a completion with no genesis-layer content, is dropped.

Proto-integer — a unary tally: length only, no multiplicity structure, no factorisation. Completion: the natural number, acquired when the tally is indexed against the lattice. (§11.)

Proto-phase — the two-valued sign $\{+, -\}$, the toggle face of the successor. Completion: continuous phase, the sign filled in by the roots of unity that the Weyl braiding generates. (§3, §5, §12.)

Proto-pi — the bare binary relation, the arity-two toll. Completion: π, entering through $\zeta(2) = \pi^2/6$ wherever a pair is drawn — through the arity of a relation, not through a circle. (§4.)

Proto-e — the integrating bridge between the multiplicative count and the additive resistance. Completion: the base e, the natural base of the isomorphism, mediating between the exponent and its logarithm. (§4.)

The genesis bit — the first step from the boundary into the interior: 2 as a period, not a third symbol; cost $\ln 2$; the two-element case in every vocabulary. Not a proto-object but the substrate's first act, collected in §22.

Shadow — the quantitative object a genesis-layer structure casts through the filter of uniqueness and irreversibility; chosen over *projection* because a shadow's kernel — what was lost in the casting — may not be known. The paper's central instance: the prime lattice as the shadow of plain ordinals. (§4, §17.)

Originating theory, axioms, hypotheses, and direction are Alexander Pisani's; the formalisation, the computational experiments, the gradings, and the drafting were developed jointly with Claude (Anthropic). Working draft, June 2026; every computational figure re-verified in the session that produced it.